STRATEGIC DEVELOPMENT AND PROGRESSION MODEL FOR AUTONOMOUS MULTI-AGENT SEARCH AND RESCUE SWARMS

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Abstract

This work presents a mathematically grounded autonomous multi-agent Search and Rescue system for coordinated drone operations over complex three-dimensional environments. The framework integrates stochastic hybrid state estimation, LiDAR–RGB–GNSS sensor fusion, Dynamic Weighted Voronoi Tessellation, Boustrophedon Cellular Decomposition, minimum-snap trajectory generation, nonlinear model predictive control, and control barrier functions. A resilient communication architecture, stochastic exception handling, payload-aware reconfiguration, and automated survivor task allocation enable adaptive mission execution. Ground Control Station telemetry optimization further supports autonomous operation under communication, sensing, environmental, and actuator constraints.

1 AXIOMATIC FOUNDATIONS, ENVIRONMENTAL MANIFOLD, AND HYBRID DYNAMICAL SYSTEM FORMULATION

1.1 The Environmental Manifold and Safe Flight Volume

The operational domain $\mathcal{W}$ is a 10-hectare ($100{,}000\,\mathrm{m}^2$) non-planar three-dimensional manifold $\mathcal{M} \subset R^3$. The uneven terrain is modeled as a continuous, piecewise-smooth scalar field

$$h : R^2 \to R_{\geq 0}, \tag{1}$$

representing the topographical elevation.

The safe flight volume $\mathcal{V}_{\text{safe}}$ is defined as the open set

$$\mathcal{V}_{\text{safe}} = \left\{ p \in R^3 \;\middle|\; p_z > h(p_x, p_y) + z_{\min} \right\}, \tag{2}$$

where

$$p = p_x p_y p_z^T \tag{3}$$

is the position vector expressed in the inertial frame, and $z_{\min} > 0$ is the minimum terrain-clearance margin (e.g., $z_{\min} = 3\,\mathrm{m}$).

Static and dynamic obstacles, such as trees and structures, are modeled as a collection of time-varying compact subsets

$$\mathcal{O}_k(t) \subset R^3. \tag{4}$$

The strictly forbidden collision space is therefore

$$\mathcal{V}_{\mathrm{col}}(t) = \bigcup_k \mathcal{O}_k(t). \tag{5}$$

Thus, the valid configuration space for the swarm at time t is defined by

$$\mathcal{C}_{\mathrm{free}}(t) = \mathcal{V}_{\mathrm{safe}} \setminus \mathcal{V}_{\mathrm{col}}(t). \tag{6}$$

1.2 Agent State-Space and Continuous Dynamics

Let the swarm be denoted by the index set

$$\mathcal{N} = \{1, 2, 3, 4\}. \tag{7}$$

The state of drone $i \in \mathcal{N}$ is represented by the 13-component quaternion-based state vector

$$\mathbf{x}_i \in R^3 \times R^3 \times S^3 \times R^3, \tag{8}$$

with

$$\mathbf{x}_i = \mathbf{p}_i^T \mathbf{v}_i^T \mathbf{q}_i^T \omega_i^{TT}, \tag{9}$$

where $\mathrm{p}_i, \mathbf{v}_i \in R^3 denote the position and linear velocity in the inertial frame \mathrm{F}_I$,
$\mathrm{q}_i = q_w q_x q_y q_z^T \in S^3 is the unit quaternion representing the orientation from \mathrm{F}_B$ to $\mathcal{F}_I$,
$\omega_i \in R^3 denotes the angular velocity expressed in the body frame \mathrm{F}_B$.

The continuous nonlinear dynamics are governed by the Newton–Euler equations of motion: $\dot{\mathbf{p}}_i = \mathbf{v}_i$,
$\dot{\mathbf{v}}_i = \mathbf{g} + \frac{1}{m_i(t)} R(\mathbf{q}_i) 0$
0
$T_i + \frac{1}{m_i(t)} \mathbf{d}_{v,i}$,
$\dot{\mathbf{q}}_i = \frac{1}{2} \mathbf{q}_i \otimes 0$
ω_i,
$\dot{\omega}_i = I_i(t)^{-1} \left[\tau_i - \omega_i \times (I_i(t) \omega_i) + \mathbf{d}_{\omega,i} \right]$.

where

$$\mathbf{g} = 00 - 9.81 \mathrm{m\,s}^{-2}, \tag{10}$$

is the gravitational acceleration vector.

Here, $m_i(t)$ and $I_i(t)$ denote the time-varying mass and inertia tensor of drone i, respectively. The scalar T_i represents the collective thrust, while

$$\tau_i = \tau_{\text{roll},i}\tau_{\text{pitch},i}\tau_{\text{yaw},i}^T \tag{11}$$

denotes the control-torque vector.

Furthermore, $R(\mathbf{q}_i) \in \text{SO}(3)$ is the 3×3 rotation matrix induced by the quaternion $\mathbf{q}_i$, while

$$\mathbf{d}_{v,i}, \mathbf{d}_{\omega,i} \in R^3 \tag{12}$$

represent unmodeled translational and rotational aerodynamic disturbances, including wind gusts and rotor–rotor or rotor–environment interactions.

1.3 Hybrid System Formulation: Discrete State Transitions (Payload Drop)

Because the drones release payloads of mass 200 g, the overall swarm constitutes a Multi-Agent Hybrid Dynamical System. Consequently, the continuous flow dynamics are interrupted by discrete state transitions.

Let $t_{\text{drop},i}$ denote the exact time at which drone i releases its payload. The mass and inertia tensor undergo instantaneous discrete transitions according to $\text{m}_i\left(t_{\text{drop},i}^+\right) = m_i\left(t_{\text{drop},i}^-\right) - \Delta m_{\text{payload}}$,

$I_i\left(t_{\text{drop},i}^+\right) = I_i\left(t_{\text{drop},i}^-\right) - \Delta I_{\text{payload}}$, where

$$\Delta m_{\text{payload}} = 0.2\,\text{kg}, \tag{13}$$

and $\Delta I_{\text{payload}}$ denotes the inertia tensor contribution of the payload relative to the drone's center of mass immediately prior to release.

Recalibration Imperative. To prevent control divergence resulting from the abrupt change in the vehicle dynamics, the control-input vector

$$u_i = T_i\tau_i^{TT} \tag{14}$$

must be instantaneously re-scaled by the Nonlinear Model Predictive Control (NMPC) layer using the updated mass and inertia parameters

$$m_i(t^+), \qquad I_i(t^+) \tag{15}$$

at the exact payload-release instant

$$t = t_{\text{drop},i}. \tag{16}$$

1.4 Kinematic, Actuator, and Geofence Constraints

The physical limits of the 4.8 kg airframe, together with the constraints imposed by the NIDAR rulebook, define strict bounds on the state and control spaces. These constraints form the inequality constraints of the optimization problem.

Actuator Saturation. The collective thrust and control torques are constrained according to

$$T_{\min} \leq T_i \leq T_{\max}, \tag{17}$$

and

$$|\tau_{k,i}| \leq \tau_{\max}, \qquad \forall k \in \{\text{roll}, \text{pitch}, \text{yaw}\}. \tag{18}$$

Kinematic Limits. To ensure stable flight and maintain the required camera orientation and focus on the terrain, the translational velocity and attitude angles are bounded as

$$\|v_i\| \leq v_{\max}, \tag{19}$$

with

$$|\phi_i| \leq \phi_{\max}, \qquad |\theta_i| \leq \theta_{\max}. \tag{20}$$

Geofence Constraint. The swarm must remain strictly within the 10-hectare boundary polygon $\mathcal{B} \subset R^2$. Hence,

$$p_{x,i}p_{y,i} \in \mathcal{B}, \qquad \forall i \in \mathcal{N}, \quad \forall t \in [0, T_{\text{mission}}]. \tag{21}$$

2 PERCEPTION LAYER (STATE ESTIMATION & MANIFOLD MAPPING) AND DECISION LAYER (DYNAMIC SPATIAL PARTITIONING)

2.1 Perception Layer: Tightly-Coupled Sensor Fusion on Non-Planar Manifolds

To guarantee collision-free operation and accurate geotagging, each drone must maintain a highly accurate estimate of its 13-dimensional state x_i (defined in Section 1.2) despite sensor noise, model uncertainty, and uneven terrain. We employ an Error-State Extended Kalman Filter (ES-EKF) that tightly fuses measurements from the IMU at 200 Hz, LiDAR at 10 Hz, RGB camera at 30 Hz, and GNSS at 10 Hz.

1. Prediction Step: IMU Mechanization. Between measurement updates, the state estimate is propagated using the IMU measurements, namely the measured linear acceleration a_m and angular velocity ω_m, while compensating for the estimated accelerometer and gyroscope biases $\hat{b}_a$ and $\hat{b}_\omega$: $\dot{\hat{p}}_i = \hat{v}_i$,

$\dot{\hat{v}}_i = R(\hat{q}_i)\left(a_m - \hat{b}_a\right) + g,$

$\dot{\hat{q}}_i = \frac{1}{2}\hat{q}_i \otimes 0$
$\omega_m - \hat{b}_\omega.$

The error-state covariance matrix P is propagated continuously according to

$$\dot{P} = FP + PF^T + Q, \tag{22}$$

where F denotes the Jacobian of the continuous-time error-state dynamics and Q is the process-noise covariance matrix.

2. LiDAR Terrain Mapping: TSDF Formulation. To navigate the non-planar manifold $\mathcal{M}$, the LiDAR point cloud $\mathcal{P}_t$ is used not only for localization but also to construct a continuous three-dimensional representation of the surrounding terrain. A Truncated Signed Distance Field (TSDF) is employed for this purpose.

For any point $p \in R^3$, the TSDF value $f(p)$ represents the signed distance to the nearest observed surface, truncated at a maximum distance μ:

$$f(p) = \max\left(-\mu, \min\left(\mu, \|p - p_{\text{surf}}\| \, sign\left(n^T\left(p - p_{\text{surf}}\right)\right)\right)\right), \tag{23}$$

where p_{surf} denotes the nearest surface point and $n \in R^3$ is the corresponding outward surface-normal vector.

Strategic Advantage. The TSDF provides a continuous scalar field whose local gradient encodes the geometry of the reconstructed surface. In particular,

$$\nabla f(p) \tag{24}$$

provides the local surface-normal direction wherever the TSDF is differentiable. This geometric information is subsequently supplied to the NMPC controller (Section 3) to construct continuous terrain-repulsion terms, thereby avoiding the discontinuous "jumping" gradients associated with discrete occupancy representations.

3. RGB-to-3D Survivor Geotagging. The NIDAR rulebook awards 25 marks for each correctly geotagged survivor. When the RGB-based convolutional neural network (e.g., YOLOv8) detects a survivor at image-plane pixel coordinates (u, v), the two-dimensional observation must be projected into the three-dimensional inertial frame.

Using the LiDAR-derived depth map $D(u, v)$ registered to the camera frame, the corresponding three-dimensional point in the camera frame is

$$p_{\text{cam}} = D(u, v)K^{-1}uv1, \tag{25}$$

where K denotes the camera intrinsic calibration matrix.

The globally geotagged survivor position is then obtained through the estimated vehicle pose and the camera-to-body extrinsic transformation:

$$p_{\text{surv}} = \hat{p}_i + R(\hat{q}_i)p_{\text{cam}} + t_{\text{cam}}^{\text{body}}, \tag{26}$$

where $t_{\text{cam}}^{\text{body}}$ denotes the calibrated translation of the camera relative to the drone body frame.

The resulting three-dimensional survivor coordinate p_{surv} is immediately broadcast to the single Ground Control Station (GCS) through the local telemetry link.

2.2 Decision Layer: Dynamic Weighted Voronoi Tessellation (DWVT)

To optimally cover the 10-hectare operational area $\mathcal{W}$ using $N = 4$ drones, the search domain must be dynamically partitioned among the available agents. Standard Voronoi tessellation assigns regions according solely to Euclidean proximity and therefore does not account for heterogeneous vehicle capabilities, battery depletion, payload availability, or mission interruptions.

We therefore formulate a Dynamic Weighted Voronoi Tessellation (DWVT) in which each drone possesses a time-varying capacity weight.

1. Weighted Partitioning. Let $c_i(t) \in R^2$ denote the horizontal position of drone i at time t. A dynamic capacity weight

$$w_i(t) \in (0, 1] \tag{27}$$

is assigned to each drone. The search domain $\mathcal{W}$ is then partitioned into the cells $\mathcal{V}_i(t)$:

$$\mathcal{V}_i(t) = \left\{ q \in \mathcal{W} \;\middle|\; \frac{\|q - c_i(t)\|}{w_i(t)} \leq \frac{\|q - c_j(t)\|}{w_j(t)}, \quad \forall j \neq i \right\}. \tag{28}$$

Weight Dynamics. The capacity weight $w_i(t)$ is dynamically modified according to the operational state of the corresponding drone. For example, following payload deployment or when the remaining battery energy falls below 30%, the weight may be reduced according to

$$w_i(t^+) = \eta_i(t) w_i(t^-), \qquad 0 < \eta_i(t) \leq 1, \tag{29}$$

with a representative degradation factor

$$\eta_i(t) = 0.5. \tag{30}$$

The reduction of $w_i(t)$ contracts the corresponding cell $\mathcal{V}_i(t)$ and transfers portions of the unsearched domain to other drones with comparatively larger capacity weights. This provides an adaptive mechanism for load redistribution without requiring manual repartitioning of the search domain.

2. Centroidal Optimization: Lloyd's Algorithm. To minimize the aggregate traversal distance of the swarm, each drone iteratively moves toward the density-weighted centroid of its assigned cell. The optimal centroid c_i^* is defined as

$$c_i^* = \frac{\int_{\mathcal{V}_i(t)} q\,\varphi(q)\,dq}{\int_{\mathcal{V}_i(t)} \varphi(q)\,dq}, \tag{31}$$

where $\varphi(q)$ denotes the spatial search-density function.

Initially, the density is assumed to be uniform:

$$\varphi(q) = 1. \tag{32}$$

When a survivor is detected at p_{surv}, a Gaussian probability-density kernel is injected at the detected location:

$$\varphi(q) = 1 + \lambda \exp\left(-\frac{\|q - p_{\text{surv}}\|^2}{2\sigma^2}\right), \tag{33}$$

where $\lambda > 0$ controls the strength of the survivor-induced priority and $\sigma > 0$ determines the spatial concentration of the Gaussian kernel.

Consequently, the centroidal optimization increases the effective density around detected survivors, causing the corresponding Voronoi boundaries to adapt and concentrating subsequent search effort in high-priority regions.

2.3 Decision Layer: Intra-Cell Path Generation

The dynamically generated Voronoi cells $\mathcal{V}_i(t)$ may be non-convex, which complicates the construction of continuous, collision-free flight trajectories. To generate systematic coverage paths, Boustrophedon Cellular Decomposition (BCD) is applied.

Convexification. Each cell $\mathcal{V}_i(t)$ is decomposed by sweeping vertical lines through the critical points, typically corresponding to polygon vertices or topological changes. This produces a collection of geometrically simpler sub-cells

$$\mathcal{S}_i = \{\mathcal{S}_{i,k}\}_{k=1}^{K_i}, \tag{34}$$

where each $\mathcal{S}_{i,k}$ is selected to be strictly convex under the adopted decomposition.

Sweep Line Generation. Within each sub-cell $\mathcal{S}_{i,k}$, parallel sweep trajectories are generated with spacing equal to the ground footprint width of the RGB camera:

$$\Delta_{\text{sweep}} = W_{\text{FOV}}, \tag{35}$$

where W_{FOV} denotes the ground-projected field-of-view width at the designated search altitude z_{search}.

The resulting Boustrophedon pattern provides systematic spatial coverage while minimizing redundant observations.

Kinematic Smoothing. Sharp direction changes at the endpoints of sweep lines may violate the inertial and attitude constraints of the 4.8 kg airframe. Accordingly, the discontinuous zig-zag corners are replaced with curvature-constrained trajectories, such as Dubins paths or minimum-curvature B-splines.

The resulting reference trajectory is required to satisfy

$$\kappa(s) \leq \kappa_{\max}, \tag{36}$$

where $\kappa(s)$ denotes the trajectory curvature as a function of arc length s, and $\kappa_{\max}$ is derived from the maximum allowable bank angle $\phi_{\max}$ and the vehicle's admissible flight speed.

2.4 Decision Layer: Dynamic Task Allocation (The Survivor Auction)

When drone i detects a survivor at p_{surv}, the swarm must determine which available drone should execute the associated 200 g payload-delivery task. This is formulated as a Dynamic Minimum-Cost Assignment Problem.

Let $\mathcal{N}$ denote the set of active drones and let $\mathcal{T}$ denote the set of detected but unserviced survivors. A cost matrix

$$C \in R^{|\mathcal{N}|\times|\mathcal{T}|} \tag{37}$$

is defined, where the cost C_{ij} represents the estimated cost for drone i to service survivor j:

$$C_{ij} = \alpha \frac{\|p_i - p_{\mathrm{surv},j}\|}{v_{\mathrm{cruise}}} + \beta \left(\frac{1}{E_i}\right) + \gamma \left(1 - \delta_{\mathrm{payload},i}\right), \tag{38}$$

where $\alpha, \beta, \gamma \in R_{\geq 0} are tuning weights corresponding to time, battery, and payload availability,$
$E_i > 0 is the remaining battery energy of drone$i,
$\delta_{\mathrm{payload},i} \in \{0,1\} indicates whether drone$i$ retains its 200\,g payload kit.$

If drone i has already released its payload, then

$$\delta_{\mathrm{payload},i} = 0, \tag{39}$$

and the corresponding assignment cost is assigned an effectively infinite penalty:

$$C_{ij} \to +\infty. \tag{40}$$

Optimization. The binary assignment matrix

$$A = [a_{ij}] \in \{0,1\}^{|\mathcal{N}|\times|\mathcal{T}|} \tag{41}$$

is obtained by solving

$$\min_{A} \sum_{i\in\mathcal{N}} \sum_{j\in\mathcal{T}} C_{ij} a_{ij}, \tag{42}$$

subject to $\sum_{i\in\mathcal{N}} a_{ij} = 1, \forall j \in \mathcal{T}$,
$\sum_{j\in\mathcal{T}} a_{ij} \leq 1, \forall i \in \mathcal{N}$.

The first constraint ensures that every detected survivor is assigned to exactly one drone, while the second ensures that each drone executes at most one active delivery task at a time.

For a square assignment problem, the optimization can be solved in $O(N^3)$ time using the Hungarian algorithm. For a decentralized swarm architecture, distributed consensus-based auction algorithms may alternatively be employed.

State Transition. Once the optimal assignment matrix A^* is obtained, the winning drone i^* immediately terminates its current Boustrophedon search trajectory. Its reference trajectory is recalculated to intercept the survivor location:

$$p_{\text{ref},i^*}(t) \leftarrow \mathcal{T}_{\text{intercept}}\left(p_{\text{surv}}\right), \tag{43}$$

after which the drone executes the payload-release maneuver and returns to its assigned search region.

Simultaneously, the DWVT mechanism of Section 2.2 updates the capacity weights $w_i(t)$, causing the search area abandoned by drone i^* to be redistributed among the remaining active drones. This enables continuous mission execution while preserving adaptive spatial coverage following survivor detection and payload deployment.

3 EXECUTION LAYER (TRAJECTORY OPTIMIZATION, NMPC RECALIBRATION, AND CONTROL BARRIER FUNCTIONS) & STOCHASTIC EXCEPTION HANDLING

3.1 Execution Layer: Minimum-Snap Trajectory Generation

To ensure smooth and energy-efficient flight while respecting the high inertia of the 4.8 kg airframes, the reference trajectory $p_{\text{ref}}(t)$ is generated by minimizing the integral of the fourth derivative of position, commonly referred to as the snap. The trajectory-generation problem is formulated as a constrained Quadratic Program (QP) using B-spline basis functions $B(t)$:

$$\min_{c} \; J = \int_0^T \left\| \frac{d^4 p(t)}{dt^4} \right\|^2 dt = c^T Q_{\text{snap}} c, \tag{44}$$

where c denotes the vector of B-spline control points and Q_{snap} is the positive-semidefinite quadratic cost matrix induced by the snap functional.

The optimization is subject to the following constraints.

Boundary Conditions. The trajectory must satisfy the prescribed initial and terminal positions:

$$p(0) = p_{\text{start}}, \qquad p(T) = p_{\text{end}}, \tag{45}$$

together with continuity and prescribed zero higher-order derivatives at the relevant waypoints.

Kinematic Limits. The translational velocity and acceleration are constrained by

$$\|\dot{p}(t)\| \leq v_{\max}, \tag{46}$$

and

$$\|\ddot{p}(t)\| \leq a_{\max}. \tag{47}$$

Terrain Avoidance. The trajectory must remain above the reconstructed terrain manifold with the prescribed minimum clearance:

$$p_z(t) \geq h\left(p_x(t), p_y(t)\right) + z_{\min}, \qquad \forall t \in [0, T], \tag{48}$$

where $h(\cdot)$ is obtained from the TSDF-based terrain representation described in Section 2.

3.2 Execution Layer: Nonlinear Model Predictive Control (NMPC) for Real-Time Recalibration

The drone tracks the generated reference trajectory $p_{\text{ref}}(t)$ using Nonlinear Model Predictive Control (NMPC). At every control cycle, operating at 50 Hz, the drone obtains an updated state estimate $\hat{x}(t)$ and solves a finite-horizon optimal control problem over the prediction horizon T_h.

The NMPC problem is formulated as

$$\min_{u(\cdot)} \int_t^{t+T_h} \left[\|x(\tau) - x_{\text{ref}}(\tau)\|_Q^2 + \|u(\tau)\|_R^2 \right] d\tau, \tag{49}$$

subject to the following constraints.

Recalibration Constraint: Initial State. The optimization is initialized using the current physically measured state:

$$x(t) = \hat{x}(t). \tag{50}$$

By forcing the prediction horizon to begin from the most recent state estimate, the NMPC controller continuously incorporates deviations caused by disturbances, payload transitions, wind, modeling errors, or navigation drift without requiring a separate explicit error-correction mechanism.

Dynamics Constraint. The predicted state trajectory must satisfy the hybrid nonlinear dynamics established in Section 1:

$$\dot{x}(\tau) = f\left(x(\tau), u(\tau)\right), \qquad \tau \in [t, t + T_h]. \tag{51}$$

The function $f(\cdot)$ incorporates the continuous Newton–Euler dynamics together with the discrete mass and inertia transitions associated with payload release.

Actuator Constraints. The control input is bounded by the physical actuator limits:

$$u_{\min} \leq u(\tau) \leq u_{\max}. \tag{52}$$

3.3 Execution Layer: Control Barrier Functions (CBFs) for Collision Avoidance

Optimization-based controllers may, under certain conditions, drive the vehicle close to environmental or inter-agent boundaries in pursuit of tracking or performance objectives. To enforce safety constraints explicitly, Control Barrier Functions (CBFs) are incorporated into the NMPC optimization problem.

1. Inter-Agent Collision Avoidance. For drones i and j, define the barrier function associated with the minimum permissible inter-agent distance d_{safe} as

$$h_{ij}(x) = \|p_i - p_j\|^2 - d_{\text{safe}}^2. \tag{53}$$

The corresponding safe set is

$$\mathcal{C}_{ij} = \{x \mid h_{ij}(x) \geq 0\}. \tag{54}$$

Forward invariance of this set can be enforced through the Lie derivative condition

$$L_f h_{ij}(x) + L_g h_{ij}(x) u_i + \alpha\left(h_{ij}(x)\right) \geq 0, \tag{55}$$

where $\alpha(\cdot)$ is an extended class-$\mathcal{K}$ function. A common linear choice is

$$\alpha(h) = \gamma h, \qquad \gamma > 0. \tag{56}$$

After arranging the terms with respect to the control input, the CBF condition can be represented in the affine form

$$A_{\text{cbf},ij} u_i \leq b_{\text{cbf},ij}\left(x_i, x_j\right), \tag{57}$$

which can be directly incorporated into the NMPC optimization problem.

2. Terrain Collision Avoidance. Similarly, define the terrain-clearance barrier function as

$$h_{\text{terr}}(x) = p_z - h(p_x, p_y) - z_{\min}. \tag{58}$$

The safe terrain set is therefore

$$\mathcal{C}_{\text{terr}} = \{x \mid h_{\text{terr}}(x) \geq 0\}. \tag{59}$$

The corresponding terrain CBF constraint is represented in the optimization problem as

$$A_{\text{terr}} u_i \leq b_{\text{terr}}(x_i). \tag{60}$$

Execution Safety Condition. If the combined NMPC optimization problem, including the hard CBF constraints, remains feasible and the CBF regularity and initial-set conditions are satisfied, forward invariance of the prescribed safe sets follows from the CBF construction. Consequently,

$$\|p_i(t) - p_j(t)\| \geq d_{\text{safe}}, \qquad \forall t \geq 0, \tag{61}$$

and

$$p_{z,i}(t) \geq h\left(p_{x,i}(t), p_{y,i}(t)\right) + z_{\min}, \qquad \forall t \geq 0. \tag{62}$$

These guarantees are conditional on model validity, constraint feasibility, accurate state estimation, and the assumptions underlying the selected CBF formulation.

3.4 Stochastic Exception Handling & Resilience Protocols

In a critical Search and Rescue environment, the autonomous system must continue to operate under communication degradation, sensor uncertainty, and actuator failures. These events are therefore modeled as stochastic transitions within the control architecture.

1. Communication Packet Loss: Local RF Mesh Degradation. Because external GSM/LTE networks are prohibited, the swarm relies on a local RF communication network. If drone i loses the state broadcast from drone j, the most recent state estimate of drone j is retained while its uncertainty region is conservatively expanded.

Let Δt denote the elapsed time since the last valid state update. The effective safety distance is defined as

$$d_{\text{safe}}^{\text{effective}} = d_{\text{safe}} + k\Delta t, \tag{63}$$

where $k > 0$ is the ghost-buffer growth coefficient.

The resulting time-varying safety radius is incorporated into the inter-agent CBF constraint, thereby increasing the separation margin as the age of the received state estimate grows.

2. Sensor Degradation: LiDAR/Visual-Odometry Drift. If the LiDAR point-cloud density decreases, for example due to dust, rain, or adverse optical conditions, the position uncertainty estimated by the Error-State Kalman Filter increases.

Let P_{pos} denote the position covariance submatrix of the ESKF covariance matrix P. A degradation event is triggered whenever

$$tr\left(P_{\text{pos}}\right) > \epsilon_{\text{LiDAR}}, \tag{64}$$

where ϵ_{LiDAR} is the prescribed uncertainty threshold.

Upon activation of the fallback mode, the NMPC weighting matrix Q is adapted to impose a stronger penalty on aggressive, high-velocity maneuvers. Simultaneously, the maximum admissible velocity is reduced:

$$v_{\max}^{\text{degraded}} = 0.5\, v_{\max}. \tag{65}$$

This controlled reduction in flight speed increases the available reaction time while the state-estimation uncertainty remains elevated.

3. Actuator Saturation and Asymmetric Payload Drop. If the payload-release mechanism becomes jammed, or if a severe crosswind causes the pitch or roll actuators to saturate, the NMPC problem may become temporarily infeasible.

In such a situation, a priority-based control-allocation mechanism is activated. The tracking objective is relaxed while the safety constraints remain hard. Formally, the optimization hierarchy satisfies

$$SafetyConstraints \succ StateTracking \succ MissionEfficiency. \tag{66}$$

Thus, if the nominal tracking problem becomes infeasible, the controller may intentionally deviate from $p_{\text{ref}}(t)$ while maintaining the CBF constraints

$$A_{\text{cbf}} u_i \leq b_{\text{cbf}}. \tag{67}$$

This establishes a fail-safe behavior in which physical separation and terrain clearance take precedence over trajectory tracking and mission speed.

4 PERCEPTION LAYER MATHEMATICS & STATE ESTIMATION

4.1 The Error-State Extended Kalman Filter (ESKF) Formulation

To avoid the singularities associated with Euler-angle parameterizations and the non-minimal representation of unit quaternions in the covariance matrix, an Error-State Extended Kalman Filter (ESKF) is employed. The nominal state is propagated using the nonlinear IMU mechanization equations, while a minimal 15-dimensional error state is estimated using a linearized Kalman filtering framework.

1. Nominal State Definition. The nominal state is defined as

$$x = p^T v^T q^T b_a^T b_\omega^{TT}, \tag{68}$$

with

$$x \in R^3 \times R^3 \times S^3 \times R^3 \times R^3, \tag{69}$$

where $p \in R^3$ denotes position, $v \in R^3$ denotes linear velocity, $q \in S^3$ is the unit quaternion representing the body-to-world orientation, and $b_a, b_\omega \in R^3$ are the accelerometer and gyroscope bias vectors, respectively.

Although the quaternion representation contributes four numerical parameters, the unit-norm constraint

$$\|q\| = 1 \tag{70}$$

means that it contains only three independent rotational degrees of freedom.

2. Error-State Definition. The ESKF represents the deviation between the true and nominal states using the minimal error state

$$\delta x = \delta p^T \delta v^T \delta\theta^T \delta b_a^T \delta b_\omega^{TT} \in R^{15}, \tag{71}$$

where $\delta\theta \in R^3$ denotes the small-angle rotation error.

For sufficiently small attitude errors, the true rotation matrix is approximated by

$$R_{\text{true}} \approx \left(I + \lfloor \delta\theta \rfloor_\times\right) R_{\text{nominal}}, \tag{72}$$

where $\lfloor \cdot \rfloor_\times$ denotes the skew-symmetric cross-product matrix.

Equivalently, the true quaternion can be represented using a multiplicative error quaternion:

$$q_{\text{true}} = \delta q \otimes q_{\text{nominal}}, \tag{73}$$

where, under the small-angle approximation,

$$\delta q \approx 1\frac{1}{2}\delta\theta. \tag{74}$$

3. Continuous-Time Prediction: IMU Mechanization. Given the raw IMU measurements a_m and ω_m, the nominal state is propagated at high frequency, e.g., 200 Hz: $\dot{p} = v$,

$\dot{v} = R(q)\,(a_m - b_a) + g$,

$\dot{q} = \frac{1}{2} q \otimes 0$
$\omega_m - b_\omega$,

$\dot{b}_a = 0$,

$\dot{b}_\omega = 0$.

The nominal bias dynamics are modeled as constant over each propagation interval. In the stochastic error-state model, however, the bias perturbations are treated as random walks and are therefore driven by the corresponding process-noise terms.

4. Discrete-Time Error-State Covariance Propagation. The continuous-time error dynamics are linearized around the current nominal state to obtain the local state-transition model. Let F_k denote the discrete error-state transition matrix over the interval $[t_k, t_{k+1}]$. The predicted covariance is propagated as

$$P_{k+1}^{-} = F_k P_k^{+} F_k^T + G_k Q_d G_k^T, \tag{75}$$

where

$$P \in R^{15\times 15} \tag{76}$$

is the error-state covariance matrix, Q_d is the discrete process-noise covariance derived from the IMU noise-density parameters, and G_k is the noise Jacobian mapping the IMU process noise into the error-state space.

The superscripts $(-)$ and $(+)$ denote the predicted and corrected covariance states, respectively.

4.2 LiDAR Odometry & Non-Planar Terrain Mapping

When a LiDAR scan $\mathcal{P}_t$ is received, typically at 10 Hz, two coupled operations are performed:

1. LiDAR scan matching for correcting the ESKF state estimate.
2. Geometric terrain reconstruction for representing the non-planar environment.

1. Scan Matching via Normal Distributions Transform (NDT). Instead of relying exclusively on point-to-point Iterative Closest Point (ICP), the proposed system employs the Normal Distributions Transform (NDT) for LiDAR registration. NDT provides improved robustness to outliers by representing the local submap as a collection of Gaussian distributions.

The local submap is partitioned into three-dimensional voxels. For voxel i, the contained LiDAR points are modeled using

$$\mathcal{N}\left(\mu_i, \Sigma_i\right), \tag{77}$$

where μ_i and Σ_i denote the mean and covariance of the points contained within voxel i.

For each point $p_j \in \mathcal{P}_t$, let $p_j(\delta x)$ denote the transformed point under the candidate pose perturbation δx. The NDT registration objective is formulated as

$$J(\delta x) = -\sum_{j\in\mathcal{P}_t} \exp\left[-\frac{1}{2}\left(p_j(\delta x) - \mu_{i(j)}\right)^T \Sigma_{i(j)}^{-1}\left(p_j(\delta x) - \mu_{i(j)}\right)\right], \tag{78}$$

where $i(j)$ denotes the voxel associated with point p_j.

The optimal pose perturbation is obtained by solving

$$\delta x_{\text{LiDAR}}^{*} = \arg\min_{\delta x} J(\delta x), \tag{79}$$

using an iterative Gauss–Newton optimization procedure.

The resulting LiDAR measurement is represented as

$$z_{\text{LiDAR}} = \delta x^*_{\text{LiDAR}}, \tag{80}$$

with associated measurement covariance

$$R_{\text{LiDAR}} = \mathcal{I}^{-1}_{\text{LiDAR}}, \tag{81}$$

where $\mathcal{I}_{\text{LiDAR}}$ denotes the information matrix obtained from the local NDT optimization.

The resulting measurement is incorporated into the ESKF update to correct the propagated pose and reduce accumulated inertial drift.

2. Truncated Signed Distance Field (TSDF) for Terrain Manifold Representation. To represent the uneven terrain as a continuous geometric manifold, a TSDF is constructed from the LiDAR observations. The TSDF is defined as a scalar field

$$f : R^3 \to R, \tag{82}$$

where $f(p)$ represents the signed distance from point $p \in R^3$ to the nearest observed surface, subject to a truncation distance μ.

For a measured surface point p_{surf} with associated surface normal n, the TSDF value at a point p is

$$f(p) = trunc_\mu \left(sign \left[n^T \left(p - p_{\text{surf}} \right) \right] \| p - p_{\text{surf}} \| \right), \tag{83}$$

where the truncation operator is defined as

$$trunc_\mu(x) = \max \left(-\mu, \min(\mu, x) \right). \tag{84}$$

The reconstructed terrain surface corresponds to the zero level set of the TSDF:

$$\mathcal{M} = \left\{ p \in R^3 \;\middle|\; f(p) = 0 \right\}. \tag{85}$$

Where the surface can locally be represented as a height field, the terrain function required by the NMPC controller is obtained implicitly from

$$f(x, y, h(x, y)) = 0. \tag{86}$$

The corresponding surface normal is obtained from the normalized TSDF gradient:

$$n(p) = \frac{\nabla f(p)}{\|\nabla f(p)\|}, \qquad \|\nabla f(p)\| \neq 0. \tag{87}$$

This differentiable geometric representation allows the controller to derive continuous terrain-clearance gradients and repulsive directions, avoiding the discontinuous gradient behavior associated with a purely discrete occupancy grid.

4.3 RGB Semantic-to-3D Projection & Geotagging

The NIDAR rulebook awards 25 marks for each correctly geotagged survivor. Consequently, a two-dimensional bounding-box detection alone is insufficient; the detected survivor must be projected into the three-dimensional inertial frame with quantified spatial uncertainty.

1. Pinhole Camera Model and Ray Casting. Let the CNN detect a survivor at the homogeneous image coordinate

$$\tilde{u} = uv1, \tag{88}$$

and let K denote the intrinsic camera calibration matrix.

The corresponding viewing ray in the camera frame is

$$r_c = K^{-1}\tilde{u}. \tag{89}$$

The ray is normalized when required:

$$\bar{r}_c = \frac{r_c}{\|r_c\|}. \tag{90}$$

2. Intersection with the TSDF Manifold. Rather than relying exclusively on a single noisy LiDAR depth value $D(u, v)$, the viewing ray is propagated through the continuous TSDF representation to determine its intersection with the reconstructed terrain or obstacle surface.

Let p_{cam} denote the camera optical center expressed in the camera frame. The ray is parameterized by

$$p(s) = p_{\text{cam}} + s\, r_c, \qquad s > 0. \tag{91}$$

The surface intersection depth s^* is obtained by solving

$$f\left(p_{\text{cam}} + s^* r_c\right) = 0. \tag{92}$$

A one-dimensional root-finding method, such as Brent's method or a secant method, can be employed to solve this equation efficiently along the ray.

The resulting three-dimensional point in the camera frame is

$$p^*_{\text{cam}} = p_{\text{cam}} + s^* r_c. \tag{93}$$

For a camera-centered coordinate system, where $p_{\text{cam}} = 0$, this simplifies to

$$p^*_{\text{cam}} = s^* r_c. \tag{94}$$

3. Transformation to the Inertial Frame: Geotagging. Let the calibrated camera-to-body extrinsic transformation be

$$T_{\text{body}}^{\text{cam}} = R_{\text{body}}^{\text{cam}} t_{\text{body}}^{\text{cam}} 01 \in SE(3), \tag{95}$$

where $R_{\text{body}}^{\text{cam}}$ and $t_{\text{body}}^{\text{cam}}$ denote the calibrated camera orientation and translation relative to the body frame.

Using the nominal ESKF state (p, q), the survivor location in the global inertial frame is

$$p_{\text{surv}} = p + R(q) \left(R_{\text{body}}^{\text{cam}} p_{\text{cam}}^{*} + t_{\text{body}}^{\text{cam}} \right). \tag{96}$$

Equivalently, using homogeneous transformations,

$$p_{\text{surv}} 1 = T_{\text{inertial}}^{\text{body}} T_{\text{body}}^{\text{cam}} p_{\text{cam}}^{*} 1, \tag{97}$$

where

$$T_{\text{inertial}}^{\text{body}} = R(q) p 01. \tag{98}$$

4. Covariance of the Geotagged Position. To quantify the confidence of each geotagged survivor position, the uncertainty is propagated from the ESKF state, camera calibration, and TSDF surface estimation.

Let

$$\Sigma_{\text{surv}} \in R^{3\times 3} \tag{99}$$

denote the covariance of the estimated survivor position. Under a first-order Taylor approximation, $\Sigma_{\text{surv}} \approx J_{\text{state}} P^{+} J_{\text{state}}^{T}$
$+ J_{\text{cam}} \Sigma_{\text{cam}} J_{\text{cam}}^{T}$
$+ J_{\text{TSDF}} \Sigma_{\text{TSDF}} J_{\text{TSDF}}^{T}$, where P^{+} is the updated ESKF covariance, while J_{state}, J_{cam}, and J_{TSDF} are the Jacobians of the geotagging transformation with respect to the vehicle state, camera calibration parameters, and TSDF-based surface estimate, respectively.

A covariance-based confidence criterion is then applied. If

$$tr\left(\Sigma_{\text{surv}}\right) > \epsilon_{\text{tag}}, \tag{100}$$

where ϵ_{tag} is the maximum permissible positional uncertainty, the Ground Control Station (GCS) flags the corresponding geotag as "Low Confidence".

The perception layer can then initiate a controlled re-observation procedure in which the responsible drone reduces its motion, reacquires the target, and performs an additional geometric scan before the geotag is accepted by the decision layer.

5 DECISION LAYER MATHEMATICS & SWARM COORDINATION

5.1 Dynamic Weighted Voronoi Tessellation (DWVT) for Spatial Partitioning

To cover the 10-hectare search area $\mathcal{W} \subset R^2$, projected onto the horizontal plane, using $N = 4$ drones, a standard Voronoi tessellation is insufficient be-

cause it assumes homogeneous agents with identical operational capabilities. In a Search and Rescue mission, a drone that has already released its 200 g payload or has depleted a significant fraction of its battery possesses a different effective operational capacity. Therefore, a Dynamic Weighted Voronoi Tessellation (DWVT) is formulated.

1. Weighted Partitioning Formulation. Let

$$c_i(t) \in R^2 \tag{101}$$

denote the horizontal position of drone i at time t. Each drone is assigned a dynamic capacity weight

$$w_i(t) \in (0, 1]. \tag{102}$$

The search area $\mathcal{W}$ is partitioned into time-varying cells $\mathcal{V}_i(t)$ according to

$$\mathcal{V}_i(t) = \left\{ q \in \mathcal{W} \;\middle|\; \frac{\|q - c_i(t)\|}{w_i(t)} \leq \frac{\|q - c_j(t)\|}{w_j(t)}, \quad \forall j \in \mathcal{N} \setminus \{i\} \right\}. \tag{103}$$

The resulting tessellation adapts continuously to changes in the operational capability of individual drones.

2. Weight Dynamics: Recalibration Logic. The capacity weight $w_i(t)$ is defined as a function of the current mission state of drone i:

$$w_i(t) = \kappa_1 \left(\frac{E_i(t)}{E_{\max}} \right) + \kappa_2 \delta_{\text{search},i}(t), \tag{104}$$

where $E_i(t)$ denotes the remaining battery energy, $E_{\max}$ is the nominal maximum battery energy, and

$$\delta_{\text{search},i}(t) \in \{0, 1\} \tag{105}$$

is a binary mission-state indicator denoting whether drone i is actively participating in the search phase.

The coefficients $\kappa_1, \kappa_2 \geq 0$ determine the relative contribution of battery availability and active-search status to the effective capacity.

When drone i is assigned to a payload-delivery task, its search-state indicator is deactivated:

$$\delta_{\text{search},i}(t) \to 0. \tag{106}$$

Consequently, its effective partition weight decreases and its assigned search region contracts. The resulting unsearched portion is redistributed among the remaining active drones through the updated DWVT boundaries.

For strict mathematical consistency, the weight should be bounded away from zero when the tessellation requires every active agent to retain a non-empty cell. Accordingly, the practical implementation may use

$$w_i(t) = \max \left(w_{\min}, \kappa_1 \frac{E_i(t)}{E_{\max}} + \kappa_2 \delta_{\text{search},i}(t) \right), \qquad w_{\min} > 0. \tag{107}$$

3. Density-Driven Centroidal Optimization: Lloyd's Algorithm. To reduce aggregate swarm transit time and concentrate search effort according to mission priorities, each drone iteratively moves toward the density-weighted centroid of its assigned cell.

The optimal centroid $c_i^*(t)$ is defined as

$$c_i^*(t) = \frac{\displaystyle\int_{\mathcal{V}_i(t)} q\,\varphi(q,t)\,dq}{\displaystyle\int_{\mathcal{V}_i(t)} \varphi(q,t)\,dq}, \tag{108}$$

where $\varphi(q,t)$ denotes the spatial search-density function.

Initially, the search density is uniform:

$$\varphi(q,0) = 1. \tag{109}$$

When a survivor is detected at position p_{surv}, a Gaussian priority kernel is introduced:

$$\varphi(q,t) = 1 + \lambda \exp\left(-\frac{\|q - p_{\text{surv}}\|^2}{2\sigma^2}\right), \tag{110}$$

where $\lambda > 0$ controls the strength of the survivor-induced search priority and $\sigma > 0$ controls its spatial extent.

The resulting density modification shifts the centroidal equilibrium toward high-priority regions and consequently concentrates search effort around detected survivors without requiring continuous centralized micromanagement.

5.2 Terrain-Aware Boustrophedon Cellular Decomposition (BCD)

The dynamically generated Voronoi cells $\mathcal{V}_i(t)$ may be non-convex polygons. Direct generation of a continuous lawnmower trajectory within such regions can produce self-intersections or invalid paths. Therefore, each cell is decomposed into geometrically simpler sub-regions while accounting for the non-planar terrain manifold.

1. Convexification via Critical-Point Slicing. The three-dimensional boundary of $\mathcal{V}_i(t)$ is first projected onto the horizontal plane. A vertical sweep line is then propagated across the resulting polygon, with decomposition events occurring at critical points such as local minima, maxima, and polygon vertices.

The resulting cell is decomposed into K_i sub-cells:

$$\mathcal{V}_i(t) = \bigcup_{k=1}^{K_i} \mathcal{S}_{i,k}(t), \tag{111}$$

where

$$\mathcal{S}_{i,k}(t) \tag{112}$$

denotes the k-th decomposed sub-cell.

The decomposition is selected such that each sub-cell satisfies the geometric requirements of the adopted BCD implementation and admits a valid continuous coverage trajectory.

2. Terrain-Corrected Sweep-Line Generation. Within each sub-cell $\mathcal{S}_{i,k}$, parallel sweep trajectories are generated. Because the terrain is represented by the non-planar manifold $\mathcal{M}$ with a local slope angle

$$\alpha(q), \tag{113}$$

the effective ground footprint of the downward-looking camera varies spatially.

Let FOV_h denote the horizontal camera field of view and let z_{rel} denote the relative clearance between the drone and the local terrain surface. The terrain-corrected ground footprint is approximated by

$$W_{\mathrm{ground}}(q) = 2z_{\mathrm{rel}} \tan\left(\frac{FOV_h}{2}\right) \cos\left(\alpha(q)\right). \tag{114}$$

The sweep-line spacing Δy is dynamically adjusted according to

$$\Delta y \leq W_{\mathrm{ground}}(q)\left(1 - \epsilon_{\mathrm{overlap}}\right), \tag{115}$$

where

$$0 < \epsilon_{\mathrm{overlap}} < 1 \tag{116}$$

is the prescribed overlap margin.

The spatially varying sweep spacing compensates for changes in terrain slope and camera footprint, reducing the probability of uncovered regions during systematic search.

3. Kinematic Smoothing: Dubins Paths. At the boundaries of each sub-cell $\mathcal{S}_{i,k}$, the drone must transition between adjacent sweep lines. A direct 180° change in heading is incompatible with the finite turning capability and inertial limitations of the 4.8 kg airframe.

The sharp corners are therefore replaced by Dubins trajectories whose minimum turning radius is determined by the vehicle's cruise velocity and maximum permissible bank angle:

$$R_{\min} = \frac{v_{\mathrm{cruise}}^2}{g \tan(\phi_{\max})}, \tag{117}$$

where $\phi_{\max}$ is the maximum admissible roll angle and g is the gravitational acceleration.

The resulting reference trajectory $p_{\mathrm{ref}}(t)$ is constructed to be at least C^1 continuous:

$$p_{\mathrm{ref}}(t) \in C^1, \tag{118}$$

ensuring continuity of position and velocity and reducing the likelihood of actuator saturation in the subsequent NMPC layer.

5.3 Centralized Task Allocation: The Survivor Auction

The NIDAR architecture mandates a single Ground Control Station (GCS). Although inter-drone communication is performed through the local communication network, the GCS serves as the centralized computational hub for mission-level task allocation.

When drone i detects a survivor at $p_{\text{surv},j}$, the three-dimensional geotag is transmitted to the GCS. The GCS then solves a Minimum-Cost Assignment Problem to determine which available drone should execute the corresponding 200 g payload-delivery task.

1. Cost Matrix Formulation. Let $\mathcal{N}$ denote the set of active drones and $\mathcal{T}$ denote the set of detected but unserviced survivors. The assignment cost matrix is defined as

$$C \in R^{|\mathcal{N}|\times|\mathcal{T}|}. \tag{119}$$

The cost C_{ij} associated with assigning drone i to survivor j is formulated as the multi-objective function $\mathrm{C}_{ij} = \underbrace{\frac{\|p_i - p_{\text{surv},j}\|}{v_{\text{cruise}}}}_{Time} + \underbrace{\lambda_1\left(1 - \frac{E_i}{E_{\max}}\right)}_{BatteryPenalty}$ $+ \underbrace{\lambda_2\left(1 - \delta_{\text{payload},i}\right)}_{PayloadConstraint} + \underbrace{\lambda_3 \frac{Area\left(\mathcal{V}_i \setminus \mathcal{V}_i^{\text{new}}\right)}{Area(\mathcal{W})}}_{SearchAbandonmentPenalty}$, where

$$\delta_{\text{payload},i} \in \{0, 1\} \tag{120}$$

indicates whether drone i still possesses its 200 g payload.

If drone i has already released its payload,

$$\delta_{\text{payload},i} = 0, \tag{121}$$

and the drone is disqualified from payload-delivery assignments by setting

$$C_{ij} \to +\infty. \tag{122}$$

The final term penalizes assignments that cause a large fraction of the drone's previously allocated search region to become temporarily uncovered.

2. Optimization: Hungarian Algorithm. The binary assignment matrix is defined as

$$A = [a_{ij}] \in \{0, 1\}^{|\mathcal{N}|\times|\mathcal{T}|}. \tag{123}$$

The optimal assignment is obtained by solving

$$\min_A \sum_{i\in\mathcal{N}} \sum_{j\in\mathcal{T}} C_{ij} a_{ij}, \tag{124}$$

subject to $\sum_{i\in\mathcal{N}} a_{ij} = 1, \forall j \in \mathcal{T}$,

$\sum_{j\in\mathcal{T}} a_{ij} \leq 1, \forall i \in \mathcal{N}$.

The first constraint ensures that each survivor receives exactly one active delivery assignment, while the second prevents an individual drone from simultaneously accepting multiple delivery tasks.

For a square assignment problem, the Hungarian algorithm (Munkres algorithm) has computational complexity

$$O(N^3). \tag{125}$$

For the proposed architecture with $N = 4$ drones, the combinatorial optimization problem is sufficiently small for real-time execution at the GCS.

Once the optimal assignment matrix A^* is obtained, the winning drone i^* terminates its current Boustrophedon trajectory and recomputes a minimum-snap reference trajectory toward $p_{\text{surv},j}$. The DWVT mechanism simultaneously updates the capacity weights $w_i(t)$, thereby redistributing the temporarily abandoned search region among the remaining active drones.

5.4 Topological Consensus & Boundary Hysteresis

A potential failure mode of Voronoi-based multi-agent coordination is boundary oscillation. If two drones operate near a Voronoi boundary, small state-estimation errors or GNSS/ESKF fluctuations may repeatedly change their cell assignments. Such oscillations can result in unnecessary trajectory changes and increased energy consumption.

To suppress this behavior, a hysteresis mechanism is introduced.

1. Hysteresis Band Formulation. Let $\partial\mathcal{V}_{ij}$ denote the shared boundary between the cells of drones i and j. A drone is not immediately reassigned when its estimated position crosses the nominal boundary. Instead, a reassignment is permitted only after the drone penetrates the neighboring region by a prescribed hysteresis distance $\epsilon_h > 0$.

Formally, if

$$p_i \in \mathcal{V}_j, \tag{126}$$

the reassignment condition is

$$dist\,(p_i, \partial\mathcal{V}_{ij}) > \epsilon_h. \tag{127}$$

This creates a finite hysteresis band around the nominal cell boundary and prevents rapid assignment switching caused by small estimation perturbations.

2. Velocity Damping at Boundaries. In addition to topological hysteresis, the reference velocity is reduced as the drone approaches a Voronoi boundary. Let

$$d_i = dist\,(p_i, \partial\mathcal{V}_i) \tag{128}$$

denote the distance from drone i to its relevant cell boundary. The reference speed is defined as

$$v_{\text{ref}}(p_i) = v_{\max} \tanh(\kappa d_i), \tag{129}$$

where $\kappa > 0$ controls the rate at which the velocity decreases near the boundary.

As $d_i \to 0$, the reference velocity satisfies

$$v_{\text{ref}} \to 0, \tag{130}$$

while sufficiently far from the boundary,

$$d_i \gg \kappa^{-1} \quad \Longrightarrow \quad v_{\text{ref}} \to v_{\max}. \tag{131}$$

The combined hysteresis and velocity-damping mechanism suppresses rapid cell switching and produces smoother spatial coordination, thereby maintaining stable collaborative mission execution under state-estimation uncertainty.

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6 EXECUTION LAYER MATHEMATICS & STOCHASTIC EXCEPTION HANDLING

6.1 Execution Layer: Minimum-Snap Trajectory Generation

To ensure smooth and energy-efficient flight while respecting the inertia of the 4.8 kg airframes, the reference trajectory $p_{\text{ref}}(t)$ between waypoints generated by the Boustrophedon planner is formulated as a constrained quadratic program (QP). The trajectory is obtained by minimizing the integral of the snap, i.e., the fourth derivative of position, using B-spline basis functions $B(t)$ with control points c:

$$\min_{c} \; J_{\text{snap}} = \int_0^T \left\| \frac{d^4 p(t)}{dt^4} \right\|^2 dt = c^T Q_{\text{snap}} c. \tag{132}$$

The optimization is subject to the following constraints.

Boundary and continuity constraints. The trajectory must satisfy the prescribed initial and terminal positions,

$$p(0) = p_{\text{start}}, \qquad p(T) = p_{\text{end}}, \tag{133}$$

together with continuity of velocity, acceleration, and jerk at intermediate waypoints. These conditions provide C^3 continuity of the reference trajectory.

Kinematic constraints. The generated trajectory must respect the vehicle's velocity and acceleration limits:

$$\|\dot{p}(t)\| \le v_{\max}, \qquad \|\ddot{p}(t)\| \le a_{\max}, \qquad \forall t \in [0, T]. \tag{134}$$

Terrain-clearance constraint. Using the continuous TSDF-derived terrain representation from the perception layer, the trajectory must remain above the terrain by at least the prescribed clearance:

$$p_z(t) \geq h_{\mathrm{TSDF}}(p_x(t), p_y(t)) + z_{\min}, \qquad \forall t \in [0, T]. \tag{135}$$

Minimizing snap rather than acceleration suppresses high-frequency variations in the commanded trajectory, thereby reducing unnecessary control activity, mechanical excitation, and associated energy consumption. This is particularly relevant for heavy-lift platforms operating under a strict mission-time constraint.

6.2 Execution Layer: Nonlinear Model Predictive Control

The reference trajectory $p_{\mathrm{ref}}(t)$ is tracked using nonlinear model predictive control (NMPC). At each control cycle, operating at 50 Hz, the estimated vehicle state $\hat{x}(t)$ is used as the initial condition for a finite-horizon optimal control problem:

$$\min_{u(\cdot)} \int_t^{t+T_h} \left[\|x(\tau) - x_{\mathrm{ref}}(\tau)\|_Q^2 + \|u(\tau)\|_R^2 \right] d\tau s.t.\ x(t) = \hat{x}(t), \quad \dot{x} = f(x, u), \quad u_{\min} \leq u(\tau) \leq u_{\max}. \tag{136}$$

The initial-state constraint

$$x(t) = \hat{x}(t) \tag{137}$$

is the fundamental recalibration mechanism of the controller. Because every optimization horizon begins from the current estimated physical state, disturbances such as wind, localization drift, and discrete payload changes are incorporated into the subsequent control solution without requiring a separate disturbance-specific correction controller.

The system dynamics $f(x, u)$ are the hybrid six-degree-of-freedom Newton–Euler dynamics defined in the physical system formulation, including time-varying mass and inertia across payload-release events.

The actuator constraints are given by

$$T_{\min} \leq T(\tau) \leq T_{\max}, \qquad \|\tau(\tau)\| \leq \tau_{\max}. \tag{138}$$

6.3 Execution Layer: Control Barrier Functions for Collision Avoidance

To prevent the tracking optimizer from sacrificing safety for trajectory performance, Control Barrier Functions (CBFs) are incorporated as hard safety constraints within the real-time optimization problem.

Inter-agent collision avoidance. For drones i and j, define the safety barrier function

$$h_{ij}(x) = \|p_i - p_j\|^2 - d_{\text{safe}}^2. \tag{139}$$

The corresponding safe set is

$$\mathcal{C}_{ij} = \{x \in \mathcal{X} \mid h_{ij}(x) \geq 0\}. \tag{140}$$

Forward invariance of this set can be enforced through the CBF condition

$$L_f h_{ij}(x) + L_g h_{ij}(x) u_i + \alpha(h_{ij}(x)) \geq 0, \tag{141}$$

where $\alpha(\cdot)$ is an extended class-$\mathcal{K}$ function, for example

$$\alpha(h) = \gamma h, \qquad \gamma > 0. \tag{142}$$

When expressed in the affine control form required by the optimization solver, the constraint becomes

$$A_{\text{CBF},ij}(x_i, x_j) u_i \leq b_{\text{CBF},ij}(x_i, x_j). \tag{143}$$

Terrain and geofence avoidance. Terrain clearance is represented by

$$h_{\text{terr}}(x) = p_z - h_{\text{TSDF}}(p_x, p_y) - z_{\min} \geq 0. \tag{144}$$

Similarly, for the geofence boundary $\partial\mathcal{B}$, define

$$h_{\text{geo}}(x) = dist(p_{xy}, \partial\mathcal{B}) \geq 0. \tag{145}$$

The complete safety-constrained controller therefore solves the tracking problem subject to the intersection of the inter-agent, terrain, and geofence safe sets.

Safety theorem. Assume that the CBF constraints remain feasible, the state estimates and dynamics satisfy the required model assumptions, the safety constraints are enforced continuously or with a sufficiently small sampling period, and the initial state lies inside the corresponding safe sets. Then forward invariance implies

$$\|p_i(t) - p_j(t)\| \geq d_{\text{safe}}, \qquad \forall i \neq j, \quad t \geq 0, \tag{146}$$

together with

$$p_z(t) \geq h_{\text{TSDF}}(p_x(t), p_y(t)) + z_{\min}, \tag{147}$$

and

$$p_{xy}(t) \in \mathcal{B}. \tag{148}$$

This guarantee is conditional on feasibility, model validity, state-estimation accuracy, and the assumptions required by the chosen CBF formulation; it does not constitute an unconditional guarantee against arbitrary wind disturbances or communication failures.

6.4 Stochastic Exception Handling & Resilience Protocols

The autonomous Search and Rescue architecture models critical failures as stochastic transitions and applies deterministic fallback policies without requiring manual intervention.

Exception 1: Communication packet loss. The drones communicate through a local RF mesh. If drone i loses the state broadcast from drone j, the last received state is retained while the uncertainty associated with the relative position increases.

Let $\Delta t_{\text{last_rx}}$ denote the elapsed time since the most recent valid state packet. An effective safety distance is defined as

$$d_{\text{safe}}^{\text{eff}}(t) = d_{\text{safe}} + k_{\text{comm}} \Delta t_{\text{last_rx}}, \qquad k_{\text{comm}} > 0. \tag{149}$$

The expanded safety distance is substituted into the inter-agent CBF constraint, thereby increasing the separation margin as communication uncertainty grows. This provides a conservative response to temporary communication degradation.

Exception 2: Sensor degradation. Under dust, fog, rain, or other adverse conditions, LiDAR degradation increases the uncertainty of the estimated position. Let P_{pos} denote the position covariance extracted from the ESKF covariance matrix. A fallback is triggered when

$$tr(P_{\text{pos}}) > \epsilon_{\text{LiDAR}}. \tag{150}$$

Upon activation, the controller increases the relative penalty associated with aggressive control inputs and reduces the admissible velocity:

$$R \leftarrow R_{\text{degraded}}, \qquad v_{\max}^{\text{degraded}} = \eta_v v_{\max}, \qquad 0 < \eta_v < 1. \tag{151}$$

For the specified fallback policy, $\eta_v = 0.5$. The drone therefore reduces its operating speed as state uncertainty increases, providing additional time for estimation and control corrections.

Exception 3: Actuator saturation and asymmetric payload release. A large crosswind or asymmetric payload release may cause the NMPC problem to become infeasible because tracking requirements conflict with physical actuator limits and safety constraints.

A lexicographic safety hierarchy is therefore imposed. If the nominal optimization problem is infeasible, trajectory-tracking objectives are relaxed before safety constraints. Conceptually,

$$Q_{\text{tracking}} \to 0 \qquad \textit{while} \qquad A_{\text{CBF}} u \le b_{\text{CBF}} \quad \textit{remainshard.} \tag{152}$$

The resulting controller may deliberately deviate from the nominal reference trajectory in order to preserve collision, terrain, and geofence safety.

Exception 4: Hybrid payload-release discontinuity. At the payload-release instant t_{drop}, the vehicle mass undergoes the discrete transition

$$m(t_{\text{drop}}^{+}) = m(t_{\text{drop}}^{-}) - \Delta m_{\text{payload}}, \qquad \Delta m_{\text{payload}} = 0.2\,\text{kg}, \tag{153}$$

with a corresponding inertia transition

$$I(t_{\text{drop}}^{+}) = I(t_{\text{drop}}^{-}) - \Delta I_{\text{payload}}. \tag{154}$$

A hardware trigger from the payload-release mechanism is used to synchronize this discrete transition with the companion computer. The NMPC prediction model is updated at t_{drop} using the post-release mass and inertia.

For a hover condition, the reduction in required thrust due solely to the 0.2 kg mass loss is

$$\Delta T_{\text{hover}} = \Delta m_{\text{payload}}\, g = 0.2 \times 9.81 = 1.962\,\text{N}. \tag{155}$$

Thus, the feedforward thrust command may be adjusted by

$$T_{\text{ff}}^{+} = T_{\text{ff}}^{-} - \Delta T_{\text{hover}}, \tag{156}$$

subject to the actuator limits and the updated NMPC solution. This compensation reduces the transient altitude error associated with the payload-release discontinuity.

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7 DECISION LAYER MATHEMATICS: SPATIAL PARTITIONING & TASK ALLOCATION

7.1 Dynamic Weighted Voronoi Tessellation (DWVT) for Spatial Partitioning

To partition the 10-hectare search area $\mathcal{W} \subset R^2$ among $N = 4$ heterogeneous drones, a standard Voronoi tessellation is insufficient because it assumes equivalent agent capabilities. Payload state, remaining energy, and current search status dynamically alter the operational capacity of each vehicle. We therefore formulate a Dynamic Weighted Voronoi Tessellation (DWVT).

Weighted partitioning. Let

$$c_i(t) \in R^2 \tag{157}$$

denote the horizontal position of drone i, and let

$$w_i(t) \in (0, 1] \tag{158}$$

denote its normalized capacity weight. The search domain is partitioned according to the multiplicatively weighted distance metric:

$$\mathcal{V}_i(t) = \left\{ q \in \mathcal{W} \;\middle|\; \frac{\|q - c_i(t)\|}{w_i(t)} \leq \frac{\|q - c_j(t)\|}{w_j(t)}, \quad \forall j \in \mathcal{N} \setminus \{i\} \right\}, \tag{159}$$

where

$$\mathcal{N} = \{1, 2, \ldots, N\}. \tag{160}$$

A smaller value of w_i reduces the effective spatial influence of drone i, thereby transferring a greater portion of the search domain to drones with larger effective capacity.

Dynamic weight recalibration. The capacity weight is continuously recalculated by the GCS from the mission state:

$$w_i(t) = \kappa_1 \frac{E_i(t)}{E_{\max}} + \kappa_2 \delta_{\text{search},i}(t), \tag{161}$$

where $E_i(t)$ is the remaining battery energy, $E_{\max}$ is the nominal maximum energy, and

$$\delta_{\text{search},i}(t) \in \{0, 1\} \tag{162}$$

indicates whether drone i is actively performing the search task.

The coefficients are selected such that the resulting weight remains within the admissible range:

$$0 < w_{\min} \leq w_i(t) \leq 1. \tag{163}$$

When drone i is assigned a payload-delivery task, its search-state variable is switched according to

$$\delta_{\text{search},i}(t) \to 0. \tag{164}$$

Consequently, its effective spatial influence is reduced and the corresponding search responsibility is redistributed through the time-varying Voronoi boundaries. The remaining drones therefore absorb the portion of the domain that is no longer actively searched by the assigned vehicle.

Density-driven centroidal optimization. To reduce aggregate transit distance while concentrating search effort in regions of greater survivor probability, each drone is directed toward the density-weighted centroid of its current cell:

$$c_i^*(t) = \frac{\displaystyle\int_{\mathcal{V}_i(t)} q\, \phi(q,t)\, dq}{\displaystyle\int_{\mathcal{V}_i(t)} \phi(q,t)\, dq}. \tag{165}$$

Initially, the search-density field is uniform:

$$\phi(q, 0) = 1. \tag{166}$$

When a survivor is detected at p_{surv}, the density field is augmented by a Gaussian kernel:

$$\phi(q, t) = 1 + \lambda \exp\left(-\frac{\|q - p_{\text{surv}}\|^2}{2\sigma^2}\right), \qquad \lambda > 0,\ \sigma > 0. \tag{167}$$

This increases the centroidal objective near the detected survivor and consequently concentrates subsequent search effort around high-probability regions.

The corresponding continuous centroid-tracking law is

$$\dot{c}_i(t) = k_p \left(c_i^*(t) - c_i(t)\right), \qquad k_p > 0. \tag{168}$$

In the implemented system, this centroid motion serves as a high-level reference for the trajectory generator and is subsequently constrained by the vehicle's kinematic and dynamic limits.

7.2 Terrain-Aware Boustrophedon Cellular Decomposition and Kinematic Smoothing

The weighted Voronoi cells $\mathcal{V}_i(t)$ may be non-convex. Direct generation of a lawnmower trajectory over a non-convex polygon can produce invalid transitions, disconnected sweep segments, or unnecessary self-intersections. Therefore, each cell is decomposed into simpler subregions before trajectory generation.

Convexification via critical-point slicing. Let the sweep axis be the x-axis and define the family of vertical slicing lines

$$L(\xi) = \left\{(x, y) \in R^2 \mid x = \xi\right\}. \tag{169}$$

Critical values are identified from changes in the connectivity of $L(\xi) \cap \mathcal{V}_i(t)$, corresponding to critical vertices of the cell boundary. These critical slices decompose the cell into subcells

$$\mathcal{V}_i(t) = \bigcup_{k=1}^{K} \mathcal{S}_{i,k}(t), \qquad \mathcal{S}_{i,k} \cap \mathcal{S}_{i,\ell} \approx \quad (k \neq \ell), \tag{170}$$

where each $\mathcal{S}_{i,k}$ is selected to admit a collision-free monotone sweep. The decomposition is not required to make every subcell strictly convex; rather, its purpose is to guarantee geometrically valid sweep connectivity.

Camera footprint and sweep-line generation. For a drone operating at relative terrain clearance z_{rel} with horizontal camera field of view FOV_h, the nominal ground footprint width is

$$W_{\mathrm{FOV}} = 2z_{\mathrm{rel}} \tan\left(\frac{FOV_h}{2}\right). \tag{171}$$

The sweep-line spacing is selected according to

$$\Delta y \leq W_{\mathrm{FOV}} \left(1 - \epsilon_{\mathrm{overlap}}\right), \qquad 0 \leq \epsilon_{\mathrm{overlap}} < 1, \tag{172}$$

where $\epsilon_{\mathrm{overlap}}$ is the prescribed overlap margin. For the intended implementation,

$$\epsilon_{\mathrm{overlap}} \in [0, 0.15]. \tag{173}$$

The inequality form explicitly accounts for terrain variation, camera projection, localization uncertainty, and implementation margins. Under the assumed visibility and terrain models, sufficiently small Δy provides continuous nominal footprint coverage.

Kinematic smoothing. At the boundaries of each subcell, consecutive sweep lines require a reversal of the vehicle heading. A discontinuous 180° reversal is incompatible with the dynamic limitations of the 4.8 kg airframes. The turning segments are therefore replaced by curvature-bounded Dubins-type transitions.

For coordinated flight at approximately constant speed v_{cruise}, the minimum turning radius imposed by the maximum admissible roll angle $\phi_{\max}$ is

$$R_{\min} = \frac{v_{\mathrm{cruise}}^2}{g \tan(\phi_{\max})}. \tag{174}$$

The reference trajectory is constructed by connecting straight sweep segments with circular arcs satisfying

$$R \geq R_{\min}. \tag{175}$$

The resulting trajectory is at least C^1-continuous, providing continuous position and velocity while avoiding instantaneous changes in heading and reducing the likelihood of actuator saturation.

7.3 Dynamic Task Allocation: Survivor Auction and Minimum-Cost Assignment

The mission architecture employs a single Ground Control Station (GCS). When drone i detects a survivor at the estimated position $p_{\mathrm{surv},j}$, the corresponding three-dimensional geotag is transmitted to the GCS. The GCS then solves a minimum-cost assignment problem to select the drone responsible for payload delivery while preserving the search efficiency of the remaining swarm.

Cost matrix formulation. Let

$$\mathcal{N}_a \tag{176}$$

denote the set of active drones and

$$\mathcal{T} \tag{177}$$

the set of detected but unserviced survivors. The cost matrix is

$$C \in R^{|\mathcal{N}_a| \times |\mathcal{T}|}. \tag{178}$$

For drone i and survivor j, define

$$C_{ij} = \frac{\|p_i - p_{\text{surv},j}\|}{v_{\text{cruise}}} + \lambda_1 \left(1 - \frac{E_i}{E_{\max}}\right) + \lambda_2 \left(1 - \delta_{\text{payload},i}\right) + \lambda_3 \frac{Area\left(\mathcal{V}_i \setminus \mathcal{V}_i^{\text{new}}\right)}{Area(\mathcal{W})}. \tag{179}$$

Here, the first term approximates the time-to-intercept, the second penalizes low remaining energy, the third encodes payload availability, and the fourth penalizes excessive abandonment of the drone's assigned search region.

The payload state is defined as

$$\delta_{\text{payload},i} \in \{0, 1\}. \tag{180}$$

If drone i has already delivered its payload, then

$$\delta_{\text{payload},i} = 0, \qquad C_{ij} = +\infty, \tag{181}$$

thereby removing the drone from the feasible assignment set.

Minimum-cost assignment. The binary assignment variables are

$$a_{ij} \in \{0, 1\}, \tag{182}$$

where $a_{ij} = 1$ indicates that drone i is assigned to survivor j. The optimization problem is

$$\min_{\mathbf{A}} \quad \sum_{i \in \mathcal{N}_a} \sum_{j \in \mathcal{T}} C_{ij} a_{ij} subject to \quad \sum_{i \in \mathcal{N}_a} a_{ij} = 1, \forall j \in \mathcal{T}, \sum_{j \in \mathcal{T}} a_{ij} \leq 1, \forall i \in \mathcal{N}_a, a_{ij} \in \{0, 1\}. \tag{183}$$

The first constraint ensures that every serviceable survivor is assigned to exactly one drone, while the second prevents any drone from simultaneously accepting multiple payload-delivery tasks.

For the square assignment case, the Hungarian (Munkres) algorithm provides an $O(N^3)$ solution. With $N = 4$, the combinatorial problem is sufficiently small for real-time execution at the GCS, subject to the implementation and communication overhead.

State transition and swarm recalculation. Let

$$\mathbf{A}^* = \arg\min_{\mathbf{A}} \sum_{i,j} C_{ij} a_{ij} \tag{184}$$

denote the optimal assignment. If

$$a^*_{i^* j^*} = 1, \tag{185}$$

then drone i^* immediately transitions from the search state to the service state. Its reference trajectory is regenerated to intercept p_{surv,j^*}, execute the 200 g payload release, and subsequently return to an admissible search trajectory.

Simultaneously, the GCS updates the DWVT state. During the service interval,

$$\delta_{\mathrm{search},i^*} \to 0, \tag{186}$$

and its effective capacity weight is reduced subject to the lower bound in Eq. eq:dwvt$_w eight_b ounds. The corresponding Voronoi boundaries therefore move continuously, redistributi$

Following payload delivery, the drone transitions back to the search state according to

$$\delta_{\mathrm{search},i^*} \to 1, \tag{187}$$

but its reduced payload capability remains encoded in the updated mission-state variables and task-allocation constraints. “‘

“‘latex

8 EXECUTION LAYER MATHEMATICS

8.1 Minimum-Snap Trajectory Generation via Constrained Quadratic Programming

The Boustrophedon planner produces a sequence of discrete waypoints

$$\mathcal{W}_{\mathrm{wp}} = \{w_0, w_1, \ldots, w_M\} \subset R^3. \tag{188}$$

The Execution Layer connects these waypoints through a smooth, dynamically feasible reference trajectory $p_{\mathrm{ref}}(t)$ that can be tracked by the NMPC controller without saturating the actuators of the 4.8 kg airframe.

B-spline parameterization. The reference trajectory is represented using a uniform B-spline of degree $d = 4$ with N_c control points:

$$c = c_1^T c_2^T \cdots c_{N_c}^{T\ T} \in R^{3N_c}, \tag{189}$$

where $c_i \in R^3$. The trajectory is

$$p(t) = \sum_{i=1}^{N_c} c_i B_{i,d}(t), \tag{190}$$

where $B_{i,d}(t)$ denotes the i-th B-spline basis function of degree d, defined over the prescribed uniform knot vector.

The k-th derivative of the B-spline curve is

$$\frac{d^k p(t)}{dt^k} = \sum_{i=1}^{N_c-k} c_i^{(k)} B_{i,d-k}(t), \tag{191}$$

where $c_i^{(k)}$ are the control points of the k-th derivative curve. For a uniform knot spacing Δt, these derivative control points are obtained recursively through finite differences. Equivalently,

$$c_i^{(k)} = \frac{d!}{(d-k)!} \frac{1}{\Delta t^k} \sum_{j=0}^{k} (-1)^{k-j} kj c_{i+j}. \tag{192}$$

Minimum-snap objective. The trajectory-generation problem minimizes the integral of the squared snap, i.e., the fourth derivative of position:

$$J_{\text{snap}} = \int_0^T \left\| \frac{d^4 p(t)}{dt^4} \right\|^2 dt = c^T Q_{\text{snap}} c, \tag{193}$$

where

$$Q_{\text{snap}} \in R^{3N_c \times 3N_c} \tag{194}$$

is the positive semidefinite quadratic-cost matrix obtained analytically from the B-spline basis functions.

Minimizing snap produces a smooth reference trajectory with continuous lower-order derivatives, thereby reducing abrupt changes in jerk and limiting high-frequency excitation of the flight-control system. This is particularly relevant for the 4.8 kg platform, where excessive control activity increases energy consumption and mechanical stress.

Constrained quadratic program. The complete trajectory-generation problem is formulated as

$$\min_c \quad c^T Q_{\text{snap}} csubjectto \quad p(t_k) = w_k, \qquad k = 0, 1, \ldots, M, \dot{p}(0) = \dot{p}(T) = 0, \ddot{p}(0) = \ddot{p}(T) = 0. \tag{195}$$

The waypoint constraints enforce exact interpolation of the Boustrophedon-generated path. Additional continuity constraints may be imposed at intermediate waypoints when required by the mission trajectory.

Kinematic limits. The reference trajectory must satisfy the vehicle's velocity and acceleration limits:

$$\|\dot{p}(t)\|_\infty \le v_{\max}, \qquad \|\ddot{p}(t)\|_\infty \le a_{\max}. \tag{196}$$

These constraints can be enforced at collocation points t_s. For B-splines, the convex-hull property provides a convenient sufficient condition through the derivative control points:

$$\left\|c_i^{(1)}\right\| \le v_{\max}, \qquad \left\|c_i^{(2)}\right\| \le a_{\max}, \qquad \forall i. \tag{197}$$

Terrain-clearance constraints. Let $f(p)$ denote the continuous TSDF representation obtained by the perception layer. The trajectory must maintain the prescribed minimum clearance:

$$p_z(t_s) \ge h_{\mathrm{TSDF}}\left(p_x(t_s), p_y(t_s)\right) + z_{\min}, \qquad \forall t_s. \tag{198}$$

Because the TSDF-derived terrain representation is generally nonlinear, the terrain constraint can be locally linearized around the trajectory obtained in the preceding optimization iteration using the TSDF gradient ∇f. This produces a locally affine constraint compatible with the QP formulation.

Computational solution. The resulting sparse QP can be solved using an appropriate quadratic-programming solver, such as OSQP or another sparse interior-point/active-set implementation. For approximately $N_c = 50$ control points and $M \approx 20$ waypoints, the computational complexity is sufficiently small for execution during the mission setup phase, subject to the actual hardware and solver implementation.

8.2 Nonlinear Model Predictive Control for Autonomous Real-Time Recalibration

The minimum-snap trajectory $p_{\mathrm{ref}}(t)$ constitutes an open-loop reference. In operation, the physical drone may deviate from this trajectory because of wind disturbances, aerodynamic effects, localization uncertainty, and discrete changes in vehicle mass during payload release. NMPC provides the closed-loop mechanism for continuously recalculating the control inputs.

Discrete-time NMPC formulation. At each control instant

$$t_k = k\Delta t, \qquad \Delta t = 0.02\,\mathrm{s}, \tag{199}$$

the current state estimate $\hat{x}_k$, obtained from the ESKF, initializes a finite-horizon optimal-control problem with N_h prediction steps:

$$\min_{u_0,\dots,u_{N_h-1}} \quad \sum_{n=0}^{N_h-1} \left[\|x_n - x_{\mathrm{ref}}(t_k + n\Delta t)\|_Q^2 + \|u_n\|_R^2 \right] + \|x_{N_h} - x_{\mathrm{ref}}(t_k + N_h\Delta t)\|_{Q_f}^2 . \tag{200}$$

The initial-state constraint is

$$x_0 = \hat{x}_k. \tag{201}$$

This constraint forces every optimization horizon to begin from the latest physically estimated state. Consequently, disturbances that have already occurred are directly incorporated into the subsequent optimal-control solution.

Discrete dynamics. The prediction model is obtained from the hybrid six-degree-of-freedom Newton–Euler dynamics:

$$x_{n+1} = x_n + \int_{t_n}^{t_n+\Delta t} f(x(\tau), u_n)\, d\tau, \qquad n = 0, \dots, N_h - 1, \tag{202}$$

where $f(\cdot)$ is the nonlinear vehicle dynamics defined in the physical-system formulation. Numerical integration is performed using fourth-order Runge–Kutta (RK4).

Actuator constraints. The control vector is subject to the physical actuator limits:

$$T_{\min} \le T_n \le T_{\max}, \tag{203}$$

and

$$|\tau_{\mathrm{roll},n}| \le \tau_{\max}, \qquad |\tau_{\mathrm{pitch},n}| \le \tau_{\max}, \qquad |\tau_{\mathrm{yaw},n}| \le \tau_{\max}. \tag{204}$$

Weighting-matrix design. The state and control weighting matrices are selected according to the physical priorities of the platform:

$$Q = diag\left(q_p, q_p, q_p, q_v, q_v, q_v, q_q, q_q, q_q, q_\omega, q_\omega, q_\omega, 0\right), \tag{205}$$

and

$$R = diag\left(r_T, r_\tau, r_\tau, r_\tau\right). \tag{206}$$

The design follows the qualitative hierarchy

$$q_p \gg q_v \gg q_\omega, \qquad r_T \gg r_\tau, \tag{207}$$

thereby prioritizing position tracking while penalizing energetically expensive thrust variations.

Real-time computational budget. The NMPC problem is formulated as a nonlinear program and solved using sequential quadratic programming (SQP), with an implementation based on CasADi/acados. For a representative horizon of

$$N_h = 20, \tag{208}$$

the resulting problem remains sufficiently compact for real-time execution on an embedded compute platform, subject to measured solver performance and implementation-specific optimization.

8.3 Control Barrier Functions for Safety-Critical Constraints

The NMPC objective promotes tracking of $p_{\text{ref}}(t)$, but tracking optimality alone does not enforce collision avoidance. Control Barrier Functions (CBFs) are therefore incorporated as hard safety constraints.

Mathematical foundation. Let

$$h : R^n \to R \tag{209}$$

be a continuously differentiable barrier function defining the safe set

$$\mathcal{C} = \{x \in R^n \mid h(x) \geq 0\}. \tag{210}$$

For a control-affine system

$$\dot{x} = f(x) + g(x)u, \tag{211}$$

a sufficient CBF condition is

$$\dot{h}(x, u) = L_f h(x) + L_g h(x) u \geq -\alpha(h(x)), \tag{212}$$

where α is an appropriate class-$\mathcal{K}$ function. A linear choice is

$$\alpha(h) = \gamma h, \qquad \gamma > 0. \tag{213}$$

Inter-agent collision avoidance. For drones i and j, define

$$h_{ij}(x_i, x_j) = \|p_i - p_j\|^2 - d_{\text{safe}}^2. \tag{214}$$

For the specified safety margin,

$$d_{\text{safe}} = 3.0\,\text{m}. \tag{215}$$

Using

$$\dot{p}_i = v_i, \tag{216}$$

the first Lie derivative is

$$L_f h_{ij} = 2(p_i - p_j)^T (v_i - v_j). \tag{217}$$

For the translational thrust contribution,

$$L_g h_{ij} = 2(p_i - p_j)^T \frac{R(q_i)}{m_i} 001. \tag{218}$$

The resulting CBF condition is therefore

$$2(p_i - p_j)^T (v_i - v_j) \quad + 2(p_i - p_j)^T \frac{R(q_i)}{m_i} 001 T_i \geq -\gamma_{ij} \left(\|p_i - p_j\|^2 - d_{\text{safe}}^2 \right). \tag{219}$$

This constraint is affine in the thrust input T_i for fixed state estimates and can therefore be embedded directly into the real-time optimization problem:

$$A_{\text{CBF},ij} u_i \leq b_{\text{CBF},ij}(x_i, x_j). \tag{220}$$

Terrain collision avoidance. Let $f(p_i)$ denote the continuous terrain representation. The terrain-clearance barrier is defined as

$$h_{\text{terr}}(x_i) = f(p_i) - z_{\min}. \tag{221}$$

Its time derivative is

$$\dot{h}_{\text{terr}} = \nabla f(p_i)^T v_i + \nabla f(p_i)^T \frac{R(q_i)}{m_i} 001 T_i. \tag{222}$$

The CBF condition becomes

$$\nabla f(p_i)^T v_i + \nabla f(p_i)^T \frac{R(q_i)}{m_i} 001 T_i \geq -\gamma_{\text{terr}} \left(f(p_i) - z_{\min} \right). \tag{223}$$

This constraint prevents the predicted state from violating the prescribed terrain-clearance invariant, subject to the validity of the terrain model and CBF assumptions.

Geofence boundary CBF. Let the 10-hectare operational boundary be represented by a polygon $\mathcal{B}$ with edges e_k, outward normals n_k, and offsets b_k. For each boundary edge, define

$$h_{\text{geo},k}(x_i) = n_k^T p_{i,xy} - b_k. \tag{224}$$

The corresponding discrete-time safety condition is expressed as

$$n_k^T v_{i,xy} + n_k^T a_{i,xy}(u_i) \Delta t \geq -\gamma_{\text{geo}} \left(n_k^T p_{i,xy} - b_k \right). \tag{225}$$

The collection of constraints for all boundary edges prevents the predicted trajectory from crossing the admissible geofence under the assumptions of the discrete-time CBF formulation.

Safety theorem. Assume that the initial state belongs to the intersection of the safe sets, the combined NMPC–CBF problem remains feasible, the state and terrain estimates satisfy the required bounded-error assumptions, and the CBF constraints are enforced at every control update. Then the corresponding safe sets are forward invariant:

$$h_{ij}(x(t)) \geq 0, \qquad h_{\text{terr}}(x(t)) \geq 0, \qquad h_{\text{geo},k}(x(t)) \geq 0, \qquad \forall t \geq t_0. \tag{226}$$

Consequently, the controller maintains the prescribed inter-agent separation, terrain clearance, and geofence constraints under the stated assumptions.

8.4 Hybrid Discontinuity: Payload-Drop Feedforward Compensation

The mission requires delivery of a 200 g payload. At the release instant t_{drop}, the drone mass transitions from

$$m_i(t_{\text{drop}}^-) = 4.8\,\text{kg} \tag{227}$$

to

$$m_i(t_{\text{drop}}^+) = 4.6\,\text{kg}. \tag{228}$$

The corresponding mass discontinuity is

$$\Delta m_{\text{payload}} = 0.2\,\text{kg}. \tag{229}$$

If the thrust command remains unchanged across the discontinuity, the change in acceleration associated with the reduced vehicle mass can be approximated by

$$\Delta a_z = \frac{\Delta m_{\text{payload}} g}{m_i(t_{\text{drop}}^+)} = \frac{0.2 \times 9.81}{4.6} \approx 0.427\,\text{m/s}^2. \tag{230}$$

Feedforward thrust compensation. The payload-release mechanism generates a hardware trigger that is received by the companion computer. At t_{drop}, the feedforward thrust is updated according to

$$T_{\text{ff}}(t_{\text{drop}}^+) = T_{\text{ff}}(t_{\text{drop}}^-) - \Delta m_{\text{payload}} g, \tag{231}$$

giving

$$\Delta T_{\text{ff}} = 0.2 \times 9.81 = 1.962\,\text{N}. \tag{232}$$

Simultaneously, the internal NMPC model is updated:

$$m_{\text{NMPC}}(t_{\text{drop}}^+) = m_{\text{NMPC}}(t_{\text{drop}}^-) - 0.2\,\text{kg}, \tag{233}$$

Table 1: Safety-critical constraint hierarchy for infeasibility handling.

Priority	Constraint	Consequence if violated
1	Terrain CBF, $h_{\text{terr}} \geq 0$	Terrain collision / mission termination
2	Inter-agent CBF, $h_{ij} \geq 0$	Inter-agent collision / mission termination
3	Geofence CBF, $h_{\text{geo}} \geq 0$	Geofence violation
4	Actuator limits, $u_{\min} \leq u \leq u_{\max}$	Hardware saturation / damage
5	Tracking objective, $\|x - x_{\text{ref}}\|_Q^2$	Reduced trajectory-tracking performance

and

$$I_{\text{NMPC}}(t_{\text{drop}}^+) = I_{\text{NMPC}}(t_{\text{drop}}^-) - \Delta I_{\text{payload}}. \tag{234}$$

This ensures that the prediction model is synchronized with the post-release physical parameters while the feedforward term provides immediate compensation before the next NMPC update.

Delivery accuracy constraint. For a prescribed delivery accuracy region, the terminal NMPC constraint at the drop waypoint can be tightened to

$$\|p(t_{\text{drop}}) - p_{\text{surv}}\| \leq 0.5\,\text{m}, \tag{235}$$

with a low release velocity requirement

$$\|v(t_{\text{drop}})\| \leq 0.3\,\text{m/s}. \tag{236}$$

For a payload released from height z_{drop}, the idealized free-fall time is

$$t_{\text{fall}} = \sqrt{\frac{2z_{\text{drop}}}{g}}, \tag{237}$$

and the corresponding horizontal displacement due to residual horizontal velocity is approximately

$$\Delta p_{xy} \approx \|v_{xy}\|\, t_{\text{fall}}. \tag{238}$$

These relations motivate the tightened terminal position and velocity constraints at the payload-release state.

8.5 Lexicographic Constraint Relaxation for Infeasibility Handling

Under extreme disturbances, the nominal NMPC–CBF optimization problem may become infeasible. Examples include strong crosswinds, simultaneous proximity to another drone, and proximity to the geofence boundary. A solver failure without a defined fallback policy could result in unsafe behavior.

A lexicographic hierarchy is therefore imposed on the optimization objectives and constraints.

If the complete optimization problem is infeasible, the controller first sacrifices tracking performance rather than safety. In particular,

$$Q_{\text{tracking}} \to 0 \tag{239}$$

allows the vehicle to abandon its nominal reference trajectory while retaining the safety constraints.

The general optimization hierarchy can therefore be represented as

$$Safetyconstraints \succ Actuatorfeasibility \succ Trajectorytracking. \tag{240}$$

If further relaxation is required, selected soft constraints may be converted into penalized slack variables:

$$h_k(x,u) + s_k \geq 0, \qquad s_k \geq 0, \tag{241}$$

with a large penalty

$$J_{\text{slack}} = \rho_k \|s_k\|^2, \qquad \rho_k \gg 1. \tag{242}$$

The terrain and inter-agent collision constraints remain hard whenever the optimization problem admits a feasible solution satisfying them. If no physically feasible control input exists that simultaneously satisfies all hard safety constraints, no optimization formulation can provide an unconditional safety guarantee; the system must instead enter a predefined safe-state behavior such as controlled hovering or emergency landing, subject to the vehicle and environment model.

8.6 Execution Layer Summary

The Execution Layer closes the loop between the geometric search plan and the physical drone dynamics through four tightly coupled mechanisms:

1. **Minimum-snap trajectory generation** converts discrete Boustrophedon waypoints into smooth, dynamically feasible reference trajectories.
2. **NMPC recalibration** repeatedly initializes the prediction horizon from the current ESKF state and updates the control solution at 50 Hz.
3. **Control Barrier Functions** impose explicit inter-agent, terrain, and geofence safety constraints within the control optimization.
4. **Hybrid payload compensation and lexicographic fallback** handle discrete mass changes and optimization infeasibility while prioritizing physical safety over nominal trajectory tracking.

Together, these mechanisms transform the geometric decisions of the Decision Layer into a dynamically feasible and safety-constrained control policy for the multi-agent aerial system. “‘

“‘latex

9 STOCHASTIC EXCEPTION HANDLING, HYBRID AUTOMATA & RESILIENCE PROTOCOLS

9.1 The Stochastic Hybrid Automaton: Mission State Machine

To provide autonomous failsafe behavior, the drone's operational logic is modeled as a Stochastic Hybrid Automaton (SHA). Let the discrete operational mode be

$$\mu \in \{\text{SEARCH}, \text{DELIVER}, \text{DEGRADED}, \text{RTH}, \text{EMERGENCY_LAND}\}. \tag{243}$$

The continuous state x evolves according to the hybrid vehicle dynamics defined in the physical-system formulation, while transitions between discrete modes are triggered by guard conditions and corresponding reset maps:

$$\mu(t^-)[\mathcal{R}]\mathcal{G}\mu(t^+). \tag{244}$$

Here, $\mathcal{G}$ denotes the set of transition guards and $\mathcal{R}$ denotes the corresponding reset maps.

Low-battery failsafe. Let $E_i(t)$ denote the remaining energy of drone i, and let $E_{\text{req}}(p_i)$ denote the estimated energy required to reach the designated $12\,\text{ft} \times 12\,\text{ft}$ landing zone from the current position p_i. The low-battery guard is

$$\mathcal{G}_{\text{low}}: \quad E_i(t) \le E_{\text{req}}(p_i) + E_{\text{margin}}. \tag{245}$$

When this condition is satisfied, the reset map is

$$\mathcal{R}_{\text{low}}: \qquad \mu(t^+) = \text{RTH}. \tag{246}$$

The drone terminates its current search trajectory, generates a direct Minimum-Snap trajectory toward the landing zone, and updates its swarm-participation state. Its DWVT search weight is reduced toward its minimum admissible value, allowing the remaining drones to absorb the corresponding search responsibility.

Communication-loss failsafe. Let Δt_{GCS} denote the elapsed time since the most recent valid heartbeat from the GCS. Communication loss is detected when

$$\mathcal{G}_{\text{comms}}: \qquad \Delta t_{\text{GCS}} > t_{\text{timeout}}, \tag{247}$$

with a representative timeout of

$$t_{\text{timeout}} = 3\,\text{s}. \tag{248}$$

The corresponding reset is

$$\mathcal{R}_{\text{comms}}: \qquad \mu(t^+) = \text{RTH}. \tag{249}$$

During this transition, the drone ceases participation in centralized task allocation and switches to onboard LiDAR–Visual–Inertial Odometry (LVIO). It follows a precomputed terrain-aware safe corridor toward the launch zone.

9.2 Sensor Degradation and Covariance-Driven Resilience

Sensor degradation caused by dust, rain, fog, glare, or other environmental effects is represented through the uncertainty of the ESKF rather than by a purely time-based trigger.

LiDAR and visual-odometry degradation. Let P_{pos} denote the position block of the ESKF covariance matrix. The degradation guard is

$$tr\left(P_{\text{pos}}\right) > \epsilon_{\text{LiDAR}}. \tag{250}$$

When this condition is satisfied, the system transitions to the degraded operating mode:

$$\mu(t^+) = \text{DEGRADED}. \tag{251}$$

The maximum admissible velocity is dynamically reduced according to

$$v_{\max}^{\text{new}} = v_{\max} \exp\left[-\kappa tr\left(P_{\text{pos}}\right)\right], \qquad \kappa > 0. \tag{252}$$

Thus, increasing position uncertainty produces a corresponding reduction in vehicle speed.

The minimum terrain-clearance requirement is simultaneously increased:

$$z_{\min}^{\text{new}} = z_{\min} + \Delta z_{\text{unc}}, \tag{253}$$

where the uncertainty-dependent margin may be defined as

$$\Delta z_{\text{unc}} = tr\left(P_{\text{pos}}\right), \tag{254}$$

provided that the covariance quantity has been appropriately scaled to a length-compatible uncertainty measure.

GNSS multipath and dropout. Let the GNSS innovation at epoch k be

$$\nu_k = z_k^{\text{GNSS}} - \hat{z}_k^{\text{GNSS}}. \tag{255}$$

A measurement is considered inconsistent when its normalized innovation exceeds the selected statistical gate. In the simplified 3σ implementation, the trigger is

$$\|\nu_k\| > 3\sigma_{\text{GNSS}} \tag{256}$$

for N consecutive epochs.

The GNSS measurement update is then rejected, while LiDAR–Inertial Odometry continues to propagate the local state. During the outage,

$$\mathcal{P}_{\mathrm{DWVT}} = \text{frozen}, \tag{257}$$

so that global spatial partitioning is not updated from an unreliable global position estimate. The affected drone continues its local Boustrophedon sweep using relative positioning until global localization is restored.

9.3 Stochastic Survivor Confirmation via Bayesian Belief Updating

A single RGB detection is insufficient to trigger an immediate payload-delivery assignment because visual detections may contain false positives caused by shadows, debris, reflections, or partial occlusion. Survivor detection is therefore represented probabilistically.

Spatial probability grid. A two-dimensional Bayesian belief grid

$$B(x, y) = P(S_{x,y}) \tag{258}$$

is maintained over the 10-hectare search domain. The initial survivor prior is

$$P(S) = 0.01. \tag{259}$$

CNN-based Bayesian update. Suppose drone i detects a candidate survivor at p_{surv} with CNN confidence C_{cnn}. The corresponding cell probability is updated according to the simplified likelihood model

$$P(S \mid D) = \frac{C_{\mathrm{cnn}} P(S)}{C_{\mathrm{cnn}} P(S) + (1 - C_{\mathrm{cnn}})(1 - P(S))}, \tag{260}$$

where D denotes the detection event.

Temporal confirmation. The candidate must remain spatially consistent over K consecutive observations. The confirmation condition is

$$K \geq K_{\min}, \tag{261}$$

with the proposed implementation using

$$K = 10 \tag{262}$$

frames, corresponding to approximately 0.3 s at a 30 Hz camera rate.

Multi-agent spatial confirmation. If another drone j independently observes the same candidate location within tolerance radius r_{tol},

$$\|p_{\text{surv},i} - p_{\text{surv},j}\| \leq r_{\text{tol}}, \tag{263}$$

the corresponding evidence is fused to increase the confidence of the candidate.

Let the resulting fused probability be $P(S \mid D_{1:m})$. The survivor is promoted to a serviceable target only when

$$P(S \mid D_{1:m}) > \epsilon_{\text{confirm}}, \tag{264}$$

with the proposed threshold

$$\epsilon_{\text{confirm}} = 0.95. \tag{265}$$

Only after this condition is satisfied is the geotag transmitted to the GCS for execution of the survivor-assignment algorithm.

9.4 Extreme Wind Gusts and QP Infeasibility Resolution

A strong crosswind can drive the vehicle toward another drone or the geofence boundary while simultaneously exhausting the available control authority. The resulting NMPC–CBF optimization problem may therefore become infeasible.

Disturbance-observer feedforward. The controller is augmented with a nonlinear disturbance observer that estimates the lumped aerodynamic disturbance

$$\hat{d}_{\text{wind},i}. \tag{266}$$

The estimated disturbance is incorporated into the control law through an appropriate inverse input map:

$$u_{\text{total}} = u_{\text{NMPC}} + g(x)^{\dagger}\hat{d}_{\text{wind}}, \tag{267}$$

where $g(x)^{\dagger}$ denotes a suitable generalized inverse or control-allocation mapping. This allows the controller to compensate proactively for estimated wind forces.

Lexicographic safety fallback. If the optimization remains infeasible after disturbance compensation, the controller applies the following priority hierarchy:

1. **Priority 1 – Hard:** terrain and inter-agent CBF constraints.
2. **Priority 2 – Hard:** actuator constraints.

3. **Priority 3 – Soft:** geofence constraint, if a boundary violation is the only alternative to a collision or loss of the vehicle.

4. **Priority 4 – Soft:** nominal trajectory-tracking objective.

The tracking objective is therefore sacrificed first:

$$Q_{\text{tracking}} \to 0, \tag{268}$$

allowing the drone to abandon its nominal trajectory and select a dynamically feasible safety-oriented trajectory.

If the geofence constraint is also relaxed, the resulting behavior explicitly trades mission-boundary compliance against the higher-priority physical-safety constraints. This trade-off must be treated as a mission-specific emergency policy rather than as an unconditional safety guarantee.

9.5 Asymmetric Payload Drop and Center-of-Gravity Shift

The 200 g payload drop changes not only the total mass but also the vehicle's center of gravity when the payload is mounted away from the nominal center.

Inertia-tensor recalibration. Prior to payload release, the loaded vehicle state is characterized by its current inertia tensor

$$I_{\text{loaded}}. \tag{269}$$

The post-release inertia tensor is updated according to

$$I^{+} = I^{-} - \Delta I_{\text{payload}}. \tag{270}$$

The corresponding center-of-gravity displacement is represented by

$$\Delta r_{\text{cg}} = r_{\text{cg}}^{+} - r_{\text{cg}}^{-}. \tag{271}$$

Feedforward torque compensation. At the payload-release instant t_{drop}, the known center-of-gravity shift produces a change in the gravitational moment. The corresponding feedforward compensation is represented as

$$\tau_{\text{ff}} = \Delta m_{\text{payload}} \left(\Delta r_{\text{cg}} \times g \right), \tag{272}$$

with the sign and frame convention selected consistently with the vehicle's body-frame dynamics.

The compensated control input becomes

$$\tau_{\text{cmd}} = \tau_{\text{NMPC}} + \tau_{\text{ff}}, \tag{273}$$

subject to the actuator limits. This reduces the transient rotational disturbance following asymmetric payload release and helps maintain camera pointing stability during subsequent survivor detection.

9.6 Sectional Conclusion: Mathematical Mandate for Autonomous SAR

The complete architecture is organized into three coupled mathematical layers:

1. **Decision Layer:** the non-planar search environment is partitioned using Dynamic Weighted Voronoi Tessellation, while survivor-service tasks are allocated through minimum-cost assignment.
2. **Execution Layer:** B-spline minimum-snap trajectories provide dynamically smooth references; NMPC performs continuous state-based recalibration; and CBFs impose explicit safety constraints.
3. **Resilience Layer:** a Stochastic Hybrid Automaton governs operational modes and autonomous transitions under battery depletion, communication loss, sensor degradation, localization failure, wind disturbances, and payload-induced hybrid events.

The resulting system is therefore modeled as a fault-tolerant multi-agent hybrid dynamical system rather than as a collection of independently scripted vehicles. Environmental uncertainty is propagated into the state-estimation covariance, mission-state transitions modify the controller and spatial allocation, and safety-critical constraints are incorporated directly into the optimization architecture.

The mathematical objective is not to assume that every disturbance is perfectly rejectable, but to ensure that each modeled failure condition produces a deterministic, autonomous, and safety-prioritized response without requiring manual intervention.

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10 EXECUTION LAYER MATHEMATICS: TRAJECTORY OPTIMIZATION & NMPC RECALIBRATION

10.1 Minimum-Snap Trajectory Generation via Constrained Quadratic Programming

The Decision Layer, through the Boustrophedon planner, outputs a sequence of discrete three-dimensional waypoints

$$\mathcal{W}_{\text{wp}} = \{w_0, w_1, \ldots, w_M\}, \qquad w_k \in R^3.$$

For a high-inertia 4.8 kg airframe, connecting these waypoints through sharp line segments or low-order polynomial trajectories can induce large control transients, actuator saturation, and increased energy consumption. Therefore, a dynamically feasible and smooth reference trajectory

$$p_{\text{ref}}(t)$$

is generated by minimizing the integral of the squared snap, i.e., the fourth derivative of position.

B-Spline Parameterization. The trajectory is represented using a uniform B-spline of degree $d = 4$ (quartic). Let the control-point vector be

$$c = c_1^T c_2^T \cdots c_{N_c}^T{}^T \in R^{3N_c}, \qquad c_i \in R^3.$$

The trajectory is parameterized as

$$p(t) = \sum_{i=1}^{N_c} c_i B_{i,4}(t),$$

where $B_{i,4}(t)$ denotes the i-th quartic B-spline basis function.

The k-th derivative is obtained analytically from finite differences of the control points:

$$\frac{d^k p(t)}{dt^k} = \sum_{i=1}^{N_c-k} c_i^{(k)} B_{i,4-k}(t),$$

where

$$c_i^{(k)} = \frac{4!}{(4-k)!} \frac{1}{\Delta t^k} \sum_{j=0}^{k} (-1)^{k-j} kj c_{i+j}.$$

Minimum-Snap Objective. The trajectory-generation objective minimizes the squared snap over the mission segment duration T:

$$J_{\text{snap}} = \int_0^T \left\| \frac{d^4 p(t)}{dt^4} \right\|^2 dt = c^T Q_{\text{snap}} c,$$

where

$$Q_{\text{snap}} \in R^{3N_c \times 3N_c}$$

is the sparse positive-semidefinite Hessian obtained analytically from the B-spline basis functions.

Minimizing the snap produces a trajectory with smooth higher-order derivatives and reduced control transients, making it suitable as a dynamically feasible reference for the subsequent NMPC stage.

Constrained Quadratic Program. The trajectory-generation problem is formulated as the quadratic program

$$\min_c \quad c^T Q_{\text{snap}} c subject to \quad waypoint, boundary, kinematic, and terrain - clearance constraints.$$

Waypoint Interpolation and Boundary Conditions. The B-spline is constrained to interpolate all planner-generated waypoints:

$$p(t_k) = w_k, \qquad k \in \{0, \dots, M\}.$$

For stationary boundary conditions, the initial and terminal velocity and acceleration are constrained as

$$\dot{p}(0) = \dot{p}(T) = 0,$$
$$\ddot{p}(0) = \ddot{p}(T) = 0.$$

Kinematic Limits. Enforcing

$$\|\dot{p}(t)\| \leq v_{\max}$$

and

$$\|\ddot{p}(t)\| \leq a_{\max}$$

at every continuous time instant would lead to an infinite set of constraints. The B-spline convex-hull property permits sufficient finite-dimensional bounds to be imposed on the derivative control points:

$$\left\|c_i^{(1)}\right\|_\infty \leq v_{\max}, \qquad \left\|c_i^{(2)}\right\|_\infty \leq a_{\max}, \qquad \forall i.$$

These bounds convert the continuous-time constraints into a finite set of linear inequalities, while providing a conservative admissible envelope for the trajectory derivatives.

Non-Planar Terrain Clearance. Let

$$f : R^3 \to R$$

denote the TSDF obtained by the Perception Layer. The terrain surface is represented locally by the zero level set

$$f(x, y, h(x, y)) = 0.$$

A minimum vertical clearance $z_{\min} > 0$ is imposed at a set of trajectory collocation points t_s:

$$p_z(t_s) \geq h_{\text{TSDF}}(p_x(t_s), p_y(t_s)) + z_{\min}.$$

For locally represented terrain through the TSDF, the nonlinear constraint can be linearized about the previous trajectory iterate $p^{(r-1)}(t_s)$:

$$f\left(p^{(r-1)}(t_s)\right) + \nabla f\left(p^{(r-1)}(t_s)\right)^T \left[p(t_s) - p^{(r-1)}(t_s)\right] \geq z_{\min}.$$

This sequential convexification/SQP procedure permits the trajectory optimizer to account for the non-planar terrain geometry while retaining a quadratic-programming structure at each iteration.

Numerical Solution. The resulting sparse QP can be solved using an interior-point or operator-splitting solver such as OSQP. For representative configurations such as

$$N_c \approx 50, \qquad M \approx 20,$$

the computational cost is implementation- and hardware-dependent and should be established experimentally on the target companion computer rather than treated as an unconditional timing guarantee.

10.2 Nonlinear Model Predictive Control for Autonomous Real-Time Recalibration

The minimum-snap trajectory

$$p_{\text{ref}}(t)$$

is an open-loop reference. Environmental disturbances, aerodynamic effects, estimation errors, and the discrete 200 g payload-release event can cause the physical vehicle to deviate from this nominal trajectory. The NMPC therefore provides the closed-loop optimization layer that continuously recalculates the control input.

Discrete-Time NMPC Formulation. At control instant

$$t_k = k\Delta t, \qquad \Delta t = 0.02\,\text{s},$$

the Error-State Kalman Filter provides the current estimated vehicle state

$$\hat{x}_k \in R^{13}.$$

The NMPC solves a finite-horizon optimal-control problem over N_h prediction steps. For example,

$$N_h = 20$$

corresponds to a prediction horizon of

$$T_h = N_h \Delta t = 0.4\,\text{s}.$$

The optimization problem is

$$\min_{\{u_n\}_{n=0}^{N_h-1}} \quad \sum_{n=0}^{N_h-1} \left[\|x_n - x_{\text{ref}}(t_k + n\Delta t)\|_Q^2 + \|u_n\|_R^2 \right] \qquad + \|x_{N_h} - x_{\text{ref}}(t_k + N_h \Delta t)\|_{Q_f}^2 .$$

Initial-State Recalibration. The critical NMPC boundary condition is

$$x_0 = \hat{x}_k.$$

Thus, every optimization cycle begins from the vehicle's current estimated physical state rather than from the nominal reference state. Disturbances that displace the vehicle from the planned path are therefore incorporated directly into the next optimal-control problem, allowing the controller to generate a dynamically feasible correction trajectory without requiring a separate outer-loop geometric correction mechanism.

Hybrid Dynamics Integration and Payload Drop. The state evolution is constrained by the nonlinear Newton–Euler dynamics introduced in the system model:

$$x_{n+1} = x_n + \int_{t_n}^{t_n+\Delta t} f(x(\tau), u_n, m(\tau), I(\tau))\, d\tau.$$

In implementation, this continuous-time integration may be evaluated using a numerical integration scheme such as fourth-order Runge–Kutta.

At the payload-release instant t_{drop}, the vehicle mass and inertia undergo a discrete reset:

$$m^+ = m^- - \Delta m_{\text{payload}},$$
$$I^+ = I^- - \Delta I_{\text{payload}},$$

with

$$\Delta m_{\text{payload}} = 0.2\,\text{kg}.$$

The updated parameters are immediately supplied to the NMPC model, causing the subsequent optimization to account for the altered vehicle dynamics and recompute the required thrust and torque commands.

Actuator Constraints. The collective thrust and body torques are constrained according to

$$T_{\min} \le T_n \le T_{\max},$$
$$\|\tau_n\|_\infty \le \tau_{\max},$$

where

$$\tau_n = \tau_{\text{roll}} \tau_{\text{pitch}} \tau_{\text{yaw}}^T.$$

Weighting Matrix Design. The state and input weighting matrices are chosen as diagonal matrices:

$$Q = diag\left(q_p, q_p, q_p, q_v, q_v, q_v, q_q, q_q, q_q, q_\omega, q_\omega, q_\omega, 0\right),$$

and

$$R = diag\left(r_T, r_\tau, r_\tau, r_\tau\right).$$

The intended weighting hierarchy is

$$q_p \gg q_v \gg q_\omega,$$

thereby prioritizing positional tracking relative to velocity and angular-rate errors. Similarly,

$$r_T \gg r_\tau$$

places a comparatively high penalty on thrust usage or variation.

Because the quaternion representation is constrained by

$$\|q\| = 1,$$

the exact treatment of the quaternion error should be implemented consistently with the chosen NMPC state parameterization; the weighting matrix above therefore represents the intended diagonal structure rather than a universal quaternion-error metric.

Solver and Computational Budget. The NMPC problem is a nonlinear program that can be solved using Sequential Quadratic Programming (SQP), for example through CasADi or acados with a Real-Time Iteration (RTI) scheme.

For

$$N_h = 20$$

and a 13-component numerical vehicle state, the target control period is

$$\Delta t = 20\,\text{ms},$$

corresponding to a 50 Hz control frequency. The actual optimization solve time depends on solver configuration, model complexity, numerical tolerances, warm-start quality, and target hardware; therefore, any claimed 8–12 ms execution time on a Jetson Orin NX should be treated as an implementation benchmark that requires experimental validation rather than as a guaranteed bound.

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11 CONTROL BARRIER FUNCTIONS (CBFs) & STOCHASTIC EXCEPTION HANDLING

11.1 Control Barrier Functions for Safety-Critical Flight

The NMPC cost function introduced in the Execution Layer incentivizes tracking of the reference trajectory

$$p_{\text{ref}}(t).$$

However, in extreme operating conditions, an unconstrained optimizer may trade safety margins against tracking performance. To enforce explicit safety constraints, Control Barrier Functions (CBFs) are incorporated as hard constraints within the NMPC optimization problem. The resulting formulation constrains the admissible control inputs with respect to inter-agent separation, terrain clearance, and geofence boundaries.

Mathematical Foundation of CBFs. Let

$$h : R^n \to R$$

be a continuously differentiable barrier function defining the safe set

$$\mathcal{C} = \{x \in R^n \mid h(x) \geq 0\}.$$

For the control-affine system

$$\dot{x} = f(x) + g(x)u,$$

a sufficient condition for forward invariance of $\mathcal{C}$ is the existence of an extended class-$\mathcal{K}$ function $\alpha(\cdot)$ satisfying

$$\dot{h}(x,u) = L_f h(x) + L_g h(x)u \geq -\alpha(h(x)),$$

where

$$L_f h(x) = \frac{\partial h}{\partial x} f(x), \qquad L_g h(x) = \frac{\partial h}{\partial x} g(x)$$

are the Lie derivatives of h along the drift and control vector fields, respectively.

Choosing

$$\alpha(h) = \gamma h, \qquad \gamma > 0,$$

gives the linear CBF condition

$$L_f h(x) + L_g h(x) u + \gamma h(x) \geq 0.$$

When $L_g h(x)$ is affine in the control variables, this condition can be incorporated directly into the NMPC optimization problem as a linear inequality:

$$A_{\text{cbf}}(x) u \leq b_{\text{cbf}}(x).$$

11.1.1 Inter-Agent Collision Avoidance CBF

For two drones i and j, define the safety barrier using the minimum permitted inter-agent distance

$$d_{\text{safe}} = 3.0\,\text{m}.$$

The barrier function is

$$h_{ij}(x_i, x_j) = \|p_i - p_j\|^2 - d_{\text{safe}}^2.$$

The corresponding safe set is

$$\mathcal{C}_{ij} = \{(x_i, x_j) \mid h_{ij}(x_i, x_j) \geq 0\}.$$

The time derivative of the barrier is

$$\dot{h}_{ij} = 2(p_i - p_j)^T (v_i - v_j).$$

Under the translational dynamics

$$\dot{v}_i = g + \frac{1}{m_i} R(q_i) 00 T_i + \frac{1}{m_i} d_{v,i},$$

the control-dependent contribution to the barrier derivative is

$$2(p_i - p_j)^T \frac{R(q_i)}{m_i} 001 T_i.$$

Neglecting or separately bounding the disturbance contribution, the CBF condition becomes

$$2(p_i - p_j)^T (v_i - v_j) \quad + 2(p_i - p_j)^T \frac{R(q_i)}{m_i} 001 T_i \geq -\gamma_{ij} \left(\|p_i - p_j\|^2 - d_{\text{safe}}^2 \right).$$

This inequality is affine in the collective thrust T_i for fixed state variables and can therefore be incorporated into the NMPC optimization as a safety constraint.

11.1.2 Terrain Collision Avoidance CBF

Let

$$f : R^3 \to R$$

denote the TSDF representation obtained from the Perception Layer. A terrain-clearance barrier is defined as

$$h_{\text{terr}}(x_i) = f(p_i) - z_{\min},$$

where $z_{\min} > 0$ denotes the required minimum clearance under the adopted TSDF convention.

Its derivative is

$$\dot{h}_{\text{terr}} = \nabla f(p_i)^T v_i.$$

The control-dependent contribution arising through the translational acceleration is

$$\nabla f(p_i)^T \frac{R(q_i)}{m_i} 001 T_i.$$

Hence, the corresponding CBF constraint is

$$\nabla f(p_i)^T v_i + \nabla f(p_i)^T \frac{R(q_i)}{m_i} 001 T_i \geq -\gamma_{\text{terr}} \left(f(p_i) - z_{\min} \right).$$

The constraint forces the optimization to maintain a nonnegative terrain-safety barrier whenever the CBF assumptions and model remain valid.

11.1.3 Geofence Boundary CBF

Let the operational region be represented by a polygon

$$\mathcal{B} \subset R^2.$$

For each edge e_k of the polygon, let

$$n_k \in R^2$$

be an inward-pointing normal and $b_k \in R$ the corresponding boundary offset. The geofence barrier is

$$h_{\text{geo},k}(x_i) = n_k^T p_{i,xy} - b_k.$$

The safe region associated with edge k is

$$h_{\text{geo},k}(x_i) \geq 0.$$

Using a discrete-time approximation over one control interval Δt, the corresponding CBF constraint can be written as

$$n_k^T v_{i,xy} + n_k^T a_{i,xy}(T_i, q_i) \Delta t \geq -\gamma_{\text{geo}} \left(n_k^T p_{i,xy} - b_k \right).$$

This constraint is evaluated for every active polygon edge and every controlled drone.

Conditional Safety Theorem. Suppose that, at the initial time t_0,

$$h_{ij}(t_0) \geq 0, \qquad h_{\text{terr}}(t_0) \geq 0, \qquad h_{\text{geo},k}(t_0) \geq 0$$

for all relevant agent pairs and geofence edges. If the combined NMPC–CBF optimization problem remains feasible and the corresponding CBF inequalities are enforced at every control update, then the associated safe sets are forward-invariant under the assumed system model and disturbance bounds.

Thus, under these conditions,

$$h_{ij}(t) \geq 0, \qquad h_{\text{terr}}(t) \geq 0, \qquad h_{\text{geo},k}(t) \geq 0, \qquad \forall t \geq t_0.$$

The resulting formulation provides a formal safety mechanism against inter-agent collision, terrain penetration, and geofence violation. It does not constitute an unconditional guarantee when the optimization problem becomes infeasible, the state estimate is invalid, or the system operates outside the assumed model and disturbance bounds.

11.2 Stochastic Exception Handling and the Hybrid Automaton

A Search and Rescue mission operates under uncertain sensing, communication loss, wind disturbances, battery depletion, and other stochastic events. The autonomous control architecture therefore represents the mission logic as a Stochastic Hybrid Automaton (SHA), combining continuous vehicle dynamics with discrete operational modes.

The discrete mode is

$$\mu \in \{\text{SEARCH}, \text{DELIVER}, \text{DEGRADED}, \text{RTH}, \text{EMERGENCY_LAND}\}.$$

Continuous state evolution follows the vehicle dynamics, while transitions

$$\mu(t^-) \to \mu(t^+)$$

are triggered by stochastic guard conditions and associated reset maps.

11.2.1 Communication Packet Loss and Local RF Mesh Degradation

Because the system relies on a local RF communication architecture, temporary loss of state broadcasts can prevent a drone from obtaining sufficiently recent information about neighboring agents. If the state of drone j is unavailable for drone i, the nominal inter-agent CBF cannot be evaluated using a current measurement.

A conservative fallback introduces a "ghost" safety buffer around the last known position of the neighboring drone. Let

$$\Delta t_{\text{last_rx}}$$

denote the elapsed time since the last valid reception of the neighboring state. The effective separation threshold is increased according to

$$d_{\text{safe}}^{\text{eff}}(t) = d_{\text{safe}} + k_{\text{ghost}} \Delta t_{\text{last_rx}},$$

where

$$k_{\text{ghost}} > 0$$

is a design parameter with units of velocity.

The inter-agent barrier is consequently replaced by

$$h_{ij}^{\text{eff}} = \|p_i - \hat{p}_j\|^2 - \left(d_{\text{safe}}^{\text{eff}}\right)^2,$$

where $\hat{p}_j$ is the last known position of drone j.

As communication uncertainty grows, the effective safety radius increases, causing the CBF constraint to become progressively more conservative. This mechanism provides a bounded-response strategy during temporary communication blackouts, subject to the validity of the assumed motion and uncertainty-growth model.

11.2.2 Sensor Degradation and Covariance-Driven Resilience

Sensor degradation is detected through the uncertainty represented by the Error-State Kalman Filter. Let

$$P_{\text{pos}}$$

denote the position-error covariance submatrix. A degradation guard is triggered when

$$Tr\left(P_{\text{pos}}\right) > \varepsilon_{\text{LiDAR}}.$$

The system then transitions to

$$\mu = \text{DEGRADED}.$$

The maximum permissible velocity is reduced according to

$$v_{\max}^{\text{new}} = v_{\max} \exp\left[-\kappa Tr(P_{\text{pos}})\right],$$

where $\kappa > 0$ is selected such that the exponent is dimensionless.

The terrain-clearance margin is also increased as a function of position uncertainty:

$$z_{\min}^{\text{new}} = z_{\min} + \Delta z_{\text{unc}},$$

where

$$\Delta z_{\text{unc}} > 0$$

is a length-valued uncertainty margin derived from the covariance, for example through a prescribed confidence-bound mapping. The covariance trace itself is not added directly to $z_{\min}$, since

$$Tr(P_{\text{pos}})$$

has units of m^2.

The resulting behavior causes the drone to reduce its velocity and increase its terrain-clearance margin as state uncertainty grows.

11.2.3 Extreme Wind Gusts and QP Infeasibility Resolution

A sufficiently large wind disturbance can cause the combined NMPC–CBF optimization problem to become infeasible when the required safety-preserving control action exceeds the available actuator authority. A solver failure must therefore be handled explicitly rather than allowing the vehicle to retain an obsolete control command.

A lexicographic constraint hierarchy is introduced:

1. **Priority 1 – Hard safety constraints:** terrain CBFs and inter-agent CBFs remain active and are not voluntarily relaxed.
2. **Priority 2 – Hard actuator constraints:**

$$u_{\min} \leq u \leq u_{\max}.$$

3. **Priority 3 – Soft geofence constraint:** geofence constraints may be softened through nonnegative slack variables when necessary to avoid a more severe physical hazard.
4. **Priority 4 – Tracking objective:** the reference-tracking weight may be reduced toward zero so that safety and physical stabilization dominate nominal mission tracking.

A generic soft-constraint formulation is

$$h_{\text{geo},k}(x,u) + s_{\text{geo},k} \geq 0, \qquad s_{\text{geo},k} \geq 0,$$

with a corresponding slack penalty

$$J_{\text{slack}} = \rho_{\text{geo}} \sum_k s_{\text{geo},k}^2.$$

This hierarchy ensures that mission tracking is sacrificed before safety-critical constraints whenever the optimization becomes constrained. However, if no control input simultaneously satisfies the hard safety and actuator constraints, mathematical optimization alone cannot guarantee collision avoidance; the system must transition to a predefined safe-state policy such as controlled hover, emergency landing, or another validated contingency maneuver.

11.2.4 Low-Battery Return-to-Home

Let

$$E_i(t)$$

denote the remaining battery energy of drone i, and let

$$E_{\text{req}}(p_i)$$

denote the estimated energy required to return from the current position p_i to the designated landing zone. Let

$$E_{\text{margin}} > 0$$

represent the required energy reserve.

The low-battery guard condition is

$$G_{\text{low}} : \qquad E_i(t) \leq E_{\text{req}}(p_i) + E_{\text{margin}}.$$

When the guard is triggered, the reset map is

$$\mu(t^+) = \text{RTH}.$$

The drone aborts the current Boustrophedon search trajectory and generates a Minimum-Snap trajectory toward the designated landing zone. Simultaneously, its DWVT influence is reduced toward the minimum admissible weight,

$$w_i(t^+) = w_{\min}, \qquad w_{\min} > 0,$$

allowing the remaining swarm members to absorb the abandoned search region.

11.3 Hybrid Discontinuity: Payload Drop Feedforward Compensation

The mission requires delivery of a 200 g payload. At the release instant

$$t_{\text{drop}},$$

the vehicle mass changes from

$$m_i^- = 4.8\,\text{kg}$$

to

$$m_i^+ = 4.6\,\text{kg},$$

assuming the nominal loaded mass is 4.8 kg.

The mass discontinuity changes the vertical dynamics. The magnitude of the corresponding acceleration change under a simplified constant-thrust interpretation is

$$\Delta a_z = \frac{\Delta m_{\text{payload}} g}{m_i^+} = \frac{0.2 \times 9.81}{4.6} \approx 0.427\,\text{m/s}^2.$$

The actual transient depends on the controller response, thrust command, aerodynamic effects, and payload-release dynamics; therefore, the resulting altitude excursion must be established experimentally or through a validated vehicle model rather than assumed from this acceleration alone.

Feedforward Thrust Compensation. The payload-release mechanism provides a hardware trigger to the companion computer at

$$t_{\text{drop}}.$$

Immediately after release, the feedforward thrust reference is updated according to

$$T_{\text{ff}}\left(t_{\text{drop}}^{+}\right) = T_{\text{ff}}\left(t_{\text{drop}}^{-}\right) - \Delta m_{\text{payload}} g,$$

with

$$\Delta m_{\text{payload}} g = 0.2 \times 9.81 = 1.962\,\text{N}.$$

Simultaneously, the NMPC internal mass parameter is reset:

$$m_{\text{NMPC}}\left(t_{\text{drop}}^{+}\right) = m_{\text{NMPC}}\left(t_{\text{drop}}^{-}\right) - 0.2\,\text{kg}.$$

The inertia model is updated consistently:

$$I_{\text{NMPC}}\left(t_{\text{drop}}^{+}\right) = I_{\text{NMPC}}\left(t_{\text{drop}}^{-}\right) - \Delta I_{\text{payload}}.$$

The feedforward compensation provides an immediate model-consistent thrust adjustment, while the subsequent NMPC iteration incorporates the updated mass and inertia into the full nonlinear optimization. The resulting hybrid control law therefore explicitly accounts for the discontinuity introduced by payload release.

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12 GCS INTEGRATION, TELEMETRY MATHEMATICS & NIDAR SCORING OPTIMIZATION

12.1 Single-Operator GCS Architecture and Telemetry Mathematics

Rules 8.13 and 8.14 mandate a single Ground Control Station (GCS) providing a unified interface for all drones, survivor detections, delivery status, live video, and mission boundaries. Since Rule 8.4 prohibits external cellular networks, mission data are transmitted through a local RF communication architecture, such as a 900 MHz LoRa link or a 5.8 GHz digital link.

Telemetry Bandwidth Optimization. The local RF communication channel has a finite bandwidth

$$B_{\text{RF}}.$$

For example, with

$$B_{\text{RF}} = 2\,\text{Mbps},$$

simultaneously transmitting high-rate state telemetry, LiDAR data, and four HD video streams can exceed the available communication capacity.

The GCS therefore implements a priority-based data multiplexing strategy. Let

$$R_{\text{state},i}, \qquad R_{\text{video},i}, \qquad R_{\text{map},i}$$

denote the allocated data rates for state, video, and map information from drone i, respectively. The rate-allocation problem is formulated as

$$\max_{\mathbf{R}} \quad \sum_{i=1}^{4} \mathbf{w}^{\mathsf{T}} \mathbf{R}_i s.t. \quad \sum_{i=1}^{4} \mathbf{1}^{\mathsf{T}} \mathbf{R}_i \leq B_{\text{RF}}, \mathbf{R}_i \succeq \mathbf{0}, \qquad i = 1, \ldots, 4.$$

The priority weights are selected as

$$w_{\text{state}} = 10, \qquad w_{\text{map}} = 5, \qquad w_{\text{video}} = 1.$$

The state telemetry stream is assigned the highest priority and is transmitted at approximately

$$f_{\text{state}} = 20\,\text{Hz}.$$

The transmitted state packet contains the vehicle state x_i, battery information, and CBF status.

The map stream has intermediate priority and is transmitted at approximately

$$f_{\text{map}} = 1\,\text{Hz}.$$

It contains compressed TSDF or terrain-map updates.

The video stream has the lowest communication priority. H.265 hardware encoding with Region-of-Interest (ROI) compression is employed. When a drone detects a potential survivor or performs a payload-delivery operation, the GCS can allocate a larger video bitrate to that drone while reducing the bitrate and frame rate of the remaining overview streams.

Unified State Synchronization. The GCS maintains a synchronized global state matrix

$$X_{\text{GCS}}(t) = x_1(t)^T x_2(t)^T x_3(t)^T x_4(t)^T \in R^{4\times 13}.$$

Because RF packets may arrive asynchronously or out of order, each telemetry packet is associated with a timestamp

$$\tau_i.$$

The GCS reconstructs the state at a common display time t_d using timestamp-aware interpolation or short-horizon state propagation. Conceptually,

$$\hat{x}_i(t_d) = \mathcal{I}\left(x_i(\tau_i), t_d - \tau_i\right),$$

where $\mathcal{I}(\cdot)$ denotes the selected interpolation or propagation operator.

This prevents discontinuous visualization of drone positions and provides a temporally consistent representation of the swarm state, survivor locations, search coverage, and mission boundaries.

12.2 Algorithmic Optimization for NIDAR Mission Scoring

The mission scoring framework is incorporated into the system design as a set of explicit engineering objectives and guard conditions. The resulting architecture maps perception accuracy, delivery precision, search speed, and safety behavior to the corresponding mission requirements.

12.2.1 Survivor Detection and Geotagging

For a detected survivor, let

$$p_{\text{surv}} \in R^3$$

denote the estimated three-dimensional geolocation and let

$$\Sigma_{\text{surv}}$$

denote its covariance matrix.

A confidence gate is applied before a detection is promoted to a confirmed GCS survivor tag:

$$Tr\left(\Sigma_{\text{surv}}\right) \leq \varepsilon_{\text{tag}}^2.$$

If

$$Tr\left(\Sigma_{\text{surv}}\right) > \varepsilon_{\text{tag}}^2,$$

the detection remains provisional and the corresponding drone can perform additional loitering or re-observation maneuvers. This reduces the probability that a poorly localized detection is immediately committed to the mission state.

The threshold

$$\varepsilon_{\text{tag}}$$

must be selected consistently with the dimensional definition of the covariance and the organizer's actual geolocation tolerance.

12.2.2 Survivor Kit Delivery Accuracy

The delivery objective is to minimize the terminal position and velocity errors at payload release:

$$\|p(t_{\text{drop}}) - p_{\text{surv}}\| \to 0,$$

$$\|v(t_{\text{drop}})\| \to 0.$$

When the drone enters a close-proximity terminal-delivery region, for example

$$\|p - p_{\text{surv}}\| \leq 3\,\text{m},$$

the nominal trajectory controller may transition to an Image-Based Visual Servoing (IBVS) controller.

Let

$$s \in R^m$$

denote the measured image feature vector and

$$s^*$$

the desired feature configuration. The image-space control law is

$$u_{\mathrm{IBVS}} = \lambda J_v^{\dagger} \left(s^* - s \right),$$

where

$$J_v$$

is the image Jacobian,

$$J_v^{\dagger}$$

is its generalized inverse, and

$$\lambda > 0$$

is the visual-servoing gain.

The desired image feature can be selected such that the target is centered in the camera field of view. This closes the terminal control loop directly through visual measurements and reduces dependence on global positioning during the final approach.

The achievable release accuracy must be validated experimentally under the actual camera calibration, target geometry, wind conditions, estimator uncertainty, and payload-release mechanism. Consequently, a numerical bound such as

$$\| p(t_{\mathrm{drop}}) - p_{\mathrm{surv}} \| < 0.3\,\mathrm{m}$$

should be treated as a design target unless independently demonstrated through testing.

12.2.3 Fast Completion and Search-Time Optimization

The search-time objective is to cover the 10-hectare operational region within the available mission interval. Let

$$W_{\mathrm{FOV}}$$

denote the effective ground footprint width of the camera and

$$\varepsilon_{\mathrm{overlap}}$$

the prescribed fractional overlap between adjacent sweep tracks.

The maximum admissible sweep velocity based on the camera integration time t_{int} and frame rate f_{frame} is approximated by

$$v_{\mathrm{sweep}} = \frac{W_{\mathrm{FOV}} \left(1 - \varepsilon_{\mathrm{overlap}} \right)}{t_{\mathrm{int}} f_{\mathrm{frame}}}.$$

The actual admissible velocity must additionally satisfy the vehicle's kinematic, perception, image-quality, and safety constraints:

$$v_{\text{sweep}} \leq \min\left(v_{\text{perception}}, v_{\max}, v_{\text{CBF}}\right).$$

For an idealized four-drone partition with negligible transition time and perfectly balanced coverage, the theoretical coverage-time lower bound is

$$T_{\min} \geq \frac{A_{\text{mission}}}{4 W_{\text{FOV}} v_{\text{sweep}}},$$

where

$$A_{\text{mission}} = 100\,000\,\text{m}^2.$$

The inequality emphasizes that the expression represents an ideal lower bound rather than an exact mission duration. Boustrophedon turns, terrain-following, survivor servicing, communication delays, unequal cell areas, and dynamic reallocation introduce additional time.

A design target may therefore be imposed as

$$T_{\text{mission}} \leq 11\,\text{min},$$

leaving approximately 4 min of nominal margin within a 15 min mission objective.

12.3 Penalty Avoidance through Mathematical Guardrails

The mission architecture incorporates explicit constraints intended to reduce the probability of penalty-triggering events.

12.3.1 Launch and Landing Zone Constraint

Let

$$\mathcal{Z}_{\text{launch}} \subset R^2$$

denote the permitted 12 ft × 12 ft launch and landing region.

During Return-to-Home operation, the trajectory optimization problem includes the terminal condition

$$p_{xy}(t_{\text{land}}) \in \mathcal{Z}_{\text{launch}},$$

together with

$$\dot{p}(t_{\text{land}}) = 0.$$

The resulting Minimum-Snap trajectory is therefore constrained to terminate inside the designated landing region with zero terminal velocity, subject to optimization feasibility and vehicle-state constraints.

12.3.2 Autonomous Operation and Manual-Intervention Prevention

The GCS architecture is designed around autonomous mission execution. The mission-control interface can restrict operator actions to explicitly authorized high-level operations such as

[LOAD MISSION FILE, START MISSION, EMERGENCY SAFETY ABORT] .

Waypoint dragging, manual trajectory correction, and manual payload-release commands can be excluded from the normal GCS control path. This software architecture reduces the possibility of accidental low-level intervention during autonomous execution.

The emergency-abort function remains a safety mechanism and must be treated separately from ordinary mission-control operations.

12.3.3 Geofence Protection

Let

$$\mathcal{B} \subset R^2$$

denote the mission boundary loaded from the organizer-provided mission file at initialization. The corresponding geofence barrier satisfies

$$h_{\text{geo}}(x) \geq 0.$$

The geofence CBF is incorporated into the NMPC constraint hierarchy. Under nominal feasibility, the controller therefore actively modifies the commanded trajectory when the vehicle approaches the mission boundary.

If a sufficiently strong disturbance causes actuator saturation and the simultaneous satisfaction of all hard constraints becomes infeasible, the previously defined lexicographic safety policy determines the appropriate fallback behavior. A geofence violation cannot be mathematically ruled out under arbitrary disturbances if actuator authority is insufficient.

12.4 Five-Minute Setup and Mission-File Ingestion Protocol

The pre-flight sequence is organized as a deterministic state machine intended to complete initialization within the 300 s setup window.

12.4.1 Automated Pre-Flight State Machine

The initialization sequence is structured as follows:

1. $t = 0$–60 s**: System boot and sensor initialization.**

   All drones boot, initialize the IMU and ESKF, initialize LiDAR processing, and establish the local RF mesh with the GCS.

2. $t = 60$–$120\,\mathrm{s}$**: Mission-file ingestion and initial partitioning.**

   The operator loads the organizer-provided mission boundary file. The GCS computes the initial DWVT partition and generates the corresponding Boustrophedon waypoint sets for the four drones.

3. $t = 120$–$180\,\mathrm{s}$**: Automated safety and actuator self-test.**

   The GCS performs the CBF consistency checks and validates the propulsion, ESC, and payload-release interfaces. Where permitted by the physical test procedure, motors may be operated at a controlled low-thrust level to verify actuator readiness.

4. $t = 180$–$240\,\mathrm{s}$**: State-estimation and navigation validation.**

   The ESKF covariance is checked against the prescribed uncertainty threshold:

   $$Tr(P_{\mathrm{pos}}) \leq \varepsilon_{\mathrm{LiDAR}}.$$

   GNSS availability is verified where applicable, and the GCS confirms that all four drones satisfy the required pre-flight conditions.

5. $t = 240\,\mathrm{s}$**: Mission authorization.**

   Once the pre-flight state machine reports all required conditions as valid, the operator initiates autonomous execution through the

   START MISSION

   command.

The remaining interval before the 300 s limit provides operational margin for minor initialization delays, subject to the actual competition rules and hardware startup characteristics.

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13 STOCHASTIC HYBRID AUTOMATON & SENSOR DEGRADATION FALLBACKS

13.1 Mathematical Framework: The Stochastic Hybrid Automaton

To comply with NIDAR Rule 8.16 (zero manual intervention) and Rule 8.19 (mandatory failsafes), the drone's operational logic is modeled as a Stochastic Hybrid Automaton (SHA) rather than as a simple deterministic script.

Let the continuous state be

$$x \in R^{13},$$

evolving according to the Newton–Euler dynamics defined previously. Let the discrete operational mode be

$$\mu(t) \in \mathcal{M} = \{\text{NOMINAL}, \text{DEGRADED_LIDAR}, \text{DEGRADED_GPS}, \text{COMMS_LOSS}, \text{EMERGENCY}\}.$$

Transitions between modes

$$\mu(t^-) \to \mu(t^+)$$

are triggered by stochastic guard conditions

$$G_{i \to j},$$

derived from the Error-State Kalman Filter (ESKF) covariance and innovation sequences. When a guard condition is satisfied, a reset map

$$R_{i \to j}$$

instantaneously modifies the NMPC weighting matrices, kinematic limits, and CBF safety margins according to the active degradation mode.

13.2 Sensor Degradation and Fallback Logic

The NIDAR scenario specifies a severe flash-flood environment in a semi-urban settlement. Such an environment can introduce high moisture, airborne dust, debris, and structural multipath effects. Rather than relying exclusively on fixed sensor-failure thresholds, the resilience architecture uses the ESKF covariance matrix and measurement innovation sequence as mathematical indicators of estimator degradation.

13.2.1 LiDAR Degradation: Dust, Fog, and Heavy Rain

In a flash-flood environment, LiDAR point-cloud density may decrease because of water droplets, dust, and atmospheric scattering. These effects can cause scan matching through NDT or ICP to fail or generate significant measurement outliers.

Mathematical Trigger. Let

$$P_{\text{pos}} \in R^{3\times 3}$$

denote the position covariance sub-block of the ESKF covariance matrix. A LiDAR-degradation guard is defined as

$$G_{\text{LIDAR}}: \qquad Tr\left(P_{\text{pos}}\right) > \varepsilon_{\text{lidar}},$$

where

$$\varepsilon_{\text{lidar}} > 0$$

denotes the maximum acceptable position uncertainty. A representative design value is

$$\varepsilon_{\text{lidar}} = 0.5\,\text{m}^2.$$

Resilience Protocol. When

$$G_{\text{LIDAR}} = 1,$$

the reset map

$$R_{\text{LIDAR}}$$

transitions the system to

$$\mu = \text{DEGRADED_LIDAR}.$$

The following autonomous reconfigurations are performed.

Measurement Gating. The LiDAR measurement is effectively removed from the ESKF update by assigning a sufficiently large measurement-noise covariance:

$$R_{\text{lidar}} \to \infty.$$

The estimator consequently falls back to the remaining navigation sources, including optical-flow/IMU estimation for horizontal motion and barometric altitude estimation for the vertical channel.

Smoothed Kinematic Downgrade. Because the 4.8 kg airframe has significant inertia, an instantaneous reduction in the velocity limit may produce undesirable control transients. The maximum velocity is therefore reduced exponentially according to the estimated uncertainty:

$$v_{\max}^{\text{new}} = v_{\max} \exp\left[-\kappa_v \left(Tr(P_{\text{pos}}) - \varepsilon_{\text{lidar}}\right)\right],$$

where

$$\kappa_v > 0$$

is a tuning parameter.

The resulting controller progressively reduces vehicle velocity as spatial uncertainty increases.

Terrain CBF Margin Expansion. The terrain Control Barrier Function introduced previously is

$$h_{\text{terr}}(x) = f(p) - z_{\min} \geq 0.$$

During LiDAR degradation, the required clearance is increased according to

$$h_{\text{terr}}^{\text{deg}}(x) = f(p) - \left(z_{\min} + Tr(P_{\text{pos}})\right) \geq 0.$$

This modification increases the terrain-safety margin as estimator uncertainty grows.

13.2.2 GPS Multipath and Dropout Near Semi-Urban Structures

Flying near buildings or flooded structures can produce GNSS multipath, in which reflected signals introduce large errors in the reported position. If these measurements are incorporated without statistical validation, the NMPC may generate aggressive corrective commands and the global spatial partitioning may become inconsistent.

Mathematical Trigger. Let the GNSS innovation at time step k be

$$\nu_k = z_{\text{gps},k} - \hat{p}_k^-,$$

where

$$z_{\text{gps},k}$$

is the measured GNSS position and

$$\hat{p}_k^-$$

is the predicted position from the ESKF.

The corresponding innovation covariance is

$$S_k = H P_k^- H^T + R_{\text{gps}}.$$

The squared Mahalanobis distance is

$$D_{M,k}^2 = \nu_k^T S_k^{-1} \nu_k.$$

The GPS degradation guard is defined as

$$G_{\text{GPS}} : \qquad D_{M,k}^2 > \chi_{3,0.999}^2.$$

For three positional degrees of freedom, the source design specifies the approximate threshold

$$\chi_{3,0.999}^2 \approx 16.27.$$

When the Mahalanobis distance exceeds this statistical threshold, the GNSS measurement is rejected as a potential multipath or outlier measurement.

Resilience Protocol. The reset map

$$R_{\text{GPS}}$$

transitions the system to

$$\mu = \text{DEGRADED_GPS}.$$

The GPS update is disabled:

$$\mathcal{G}_{\text{GPS}} = 0,$$

and the measurement

$$z_{\text{gps}}$$

is rejected by the estimator.

The ESKF consequently relies on LiDAR–Inertial Odometry (LIO) or Visual–Inertial Odometry (VIO) for local navigation.

Swarm Coordination Halt. The Dynamic Weighted Voronoi Tessellation (DWVT) depends on globally consistent coordinates to partition the 10-hectare search region. If drone i loses reliable GNSS positioning, its global estimate can drift relative to the remaining agents.

The GCS therefore freezes the global Voronoi partition:

$$\mathcal{V}_i(t) = \mathcal{V}_i(t_{\text{freeze}}),$$

and drone i continues its local Boustrophedon sweep using relative LIO/VIO positioning.

Global task reallocation, including the Survivor Auction, is suspended until the global positioning solution satisfies the required validity criteria. This prevents abrupt global-coordinate jumps from corrupting inter-agent coordination.

13.2.3 Blind Failsafe: Total Sensor Degradation

If both LiDAR and GPS become unreliable, for example when a drone enters a dense GPS-denied structure within the operational region, the ESKF covariance can increase because the remaining inertial measurements are subject to accumulated drift.

Mathematical Trigger. The blind-state guard is defined as

$$G_{\text{BLIND}}: \qquad Tr\left(P_{\text{pos}}\right) > \varepsilon_{\text{critical}} \quad \wedge \quad \text{GPS gated}.$$

When the guard is satisfied, the reset map

$$R_{\text{BLIND}}$$

transitions the drone to

$$\mu = \text{EMERGENCY_HOVER}.$$

Emergency Hover Protocol. The drone immediately aborts the active Boustrophedon trajectory. The NMPC reference is reset to the current estimated position:

$$p_{\text{ref}}(t) = \hat{p}(t).$$

The controller then commands a smooth reduction in translational velocity:

$$\lim_{t \to t_{\text{hover}}} v(t) = 0,$$

while maintaining attitude through the IMU and altitude through the available barometric measurement.

Swarm Recalculation. The GCS marks drone i as

$$\text{BLIND}.$$

Its DWVT influence is then reduced according to

$$w_i(t^+) \to 0,$$

transferring the corresponding unsearched region to the remaining three drones.

Once the estimator uncertainty decreases below the recovery threshold,

$$Tr\left(P_{\text{pos}}\right) \leq \varepsilon_{\text{lidar}},$$

the drone may exit the emergency mode, reinitialize its local state estimate, and resume autonomous operation using its newly assigned Voronoi region.

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14 COMMUNICATION TOPOLOGY RESILIENCE & ACTUATOR SATURATION HANDLING

14.1 Communication Packet Loss & Decentralized CBF Fallback

Rules 8.4 and 8.17 prohibit GSM/LTE/5G communication. The swarm therefore relies on a local RF mesh, such as a 900 MHz LoRa or 5.8 GHz digital telemetry link, for inter-drone state broadcasting and GCS uplink.

Mathematical Threat to Inter-Agent CBFs. The inter-agent collision-avoidance barrier is defined as

$$h_{ij}(x) = \|p_i - p_j\|^2 - d_{\text{safe}}^2.$$

Evaluation of this barrier requires sufficiently recent state information from drone j. If the corresponding RF packet is lost, drone i no longer has direct access to the current state x_j.

Let

$$t_{\text{last_rx}}^{(j)}$$

denote the timestamp of the most recently received valid state broadcast from drone j. The communication-loss guard is defined as

$$G_{\text{COMMS}}: \qquad t_{\text{current}} - t_{\text{last_rx}}^{(j)} > t_{\text{timeout}},$$

where $t_{\text{timeout}} > 0$ is the maximum permissible state-information age.

A representative design value is

$$t_{\text{timeout}} = 200\,\text{ms}.$$

Ghost-Buffer Expansion. When

$$G_{\text{COMMS}} = 1,$$

drone i conservatively propagates the uncertainty associated with the unobserved motion of drone j. Let

$$\Delta t = t_{\text{current}} - t^{(j)}_{\text{last_rx}}.$$

Assuming bounded translational speed and acceleration,

$$\|v_j(t)\| \le v_{\max}, \qquad \|a_j(t)\| \le a_{\max},$$

an upper-bound ghost radius can be defined as

$$r_{\text{ghost}}(\Delta t) = v_{\max}\Delta t + \frac{1}{2}a_{\max}\Delta t^2.$$

The effective safety distance is then inflated:

$$d^{\text{eff}}_{\text{safe}}(t) = d_{\text{safe}} + r_{\text{ghost}}(\Delta t).$$

Using the last known position

$$\hat{p}_j = p_j\left(t^{(j)}_{\text{last_rx}}\right),$$

the conservative ghost barrier becomes

$$h^{\text{ghost}}_{ij} = \|p_i - \hat{p}_j\|^2 - \left(d^{\text{eff}}_{\text{safe}}\right)^2 \ge 0.$$

The inflated safety boundary becomes progressively more conservative as the age of the neighboring drone's state increases. The validity of this construction depends on the assumed bounds $v_{\max}$ and $a_{\max}$ remaining valid during the communication outage.

Onboard Autonomous Return-to-Home Trigger. If communication with the GCS is unavailable for longer than the critical timeout

$$\Delta t_{\text{GCS}} > t_{\text{critical}},$$

the onboard Stochastic Hybrid Automaton transitions autonomously to

$$\mu = \text{RTH}.$$

For example,

$$t_{\text{critical}} = 5\,\text{s}.$$

The drone abandons global Voronoi search coordination, switches to locally available LiDAR–Inertial Odometry, and follows a precomputed terrain-aware safe corridor toward the designated $12\,\text{ft} \times 12\,\text{ft}$ landing zone. The RTH transition is therefore generated onboard rather than requiring a manual return command from the GCS.

14.2 Actuator Saturation, Extreme Wind Gusts & Lexicographic QP Resolution

Severe weather conditions can introduce large lateral disturbances. For a 4.8 kg airframe, a sufficiently strong crosswind can drive the commanded pitch and roll moments toward their actuator limits.

Mathematical Threat: QP Infeasibility. Let the actuator constraints be

$$u_{\min} \leq u \leq u_{\max}.$$

During a large disturbance, the CBF constraints may demand a control action that lies outside this admissible actuator set. The resulting NMPC–CBF optimization problem may therefore become infeasible:

$$\mathcal{U}_{\mathrm{CBF}}(x) \cap \mathcal{U}_{\mathrm{act}} = .$$

A robust controller must explicitly detect this condition and transition to a predefined fallback policy rather than relying on retention of a stale control command.

14.2.1 Nonlinear Disturbance Observer

To reduce the disturbance burden placed on the NMPC optimizer, a nonlinear Disturbance Observer (DOB) estimates the lumped wind disturbance

$$\hat{d}_{\mathrm{wind}} \in R^3.$$

A representative observer dynamics is

$$\dot{\hat{d}}_{\mathrm{wind}} = -L\hat{d}_{\mathrm{wind}} + L\left[m\dot{v} - mg - R(q)00T\right],$$

where

$$L \succ 0$$

is the observer gain matrix.

The estimated disturbance can then be incorporated into the control allocation. Rather than using an ordinary matrix inverse, which may not exist for a general nonlinear or over-/under-actuated input map, the generalized inverse is used:

$$u_{\mathrm{total}} = u_{\mathrm{NMPC}} + g(x)^{\dagger}\hat{d}_{\mathrm{wind}},$$

where

$$g(x)^{\dagger}$$

denotes an appropriate generalized inverse or control-allocation operator.

The purpose of this feedforward compensation is to counteract the estimated disturbance before it produces a large tracking error, thereby reducing the corrective burden on the NMPC.

Table 2: Safety-prioritized constraint hierarchy for extreme disturbances.

Priority	Constraint	Consequence	Treatment
1	Terrain CBF	Physical collision	Hard; never voluntarily relaxed
2	Inter-agent CBF	Inter-agent collision	Hard; never voluntarily relaxed
3	Actuator limits	Hardware saturation/damage	Hard
4	Geofence CBF	Boundary violation	Soft; slack permitted
5	Tracking objective	Mission-performance degradation	Soft; reduced or removed

14.2.2 Lexicographic Constraint Resolution

If the optimization remains infeasible despite disturbance compensation, the controller applies a lexicographic constraint hierarchy.

The geofence constraint can be softened using a nonnegative slack variable:

$$h_{\text{geo}}(x, u) + \epsilon_{\text{geo}} \geq 0, \qquad \epsilon_{\text{geo}} \geq 0,$$

with a penalty

$$J_{\text{geo}} = \rho_{\text{geo}} \epsilon_{\text{geo}}^2, \qquad \rho_{\text{geo}} \gg 0.$$

The tracking objective can be progressively de-emphasized through

$$Q \to 0,$$

causing the controller to abandon nominal trajectory tracking and prioritize the remaining safety constraints.

The resulting optimization can be expressed conceptually as

$$\min \left[J_{\text{track}}, J_{\text{geo}}, J_{\text{other}} \right]_{\text{lex}},$$

subject to the hard terrain, inter-agent, and actuator constraints.

If the drone is simultaneously driven toward another drone and toward a geofence boundary, the lexicographic hierarchy prioritizes the collision-avoidance constraint over the geofence constraint. Thus, under the specified hierarchy, a geofence violation may be accepted if that is the only feasible control action that preserves the higher-priority physical safety constraint.

This is a mission-specific constraint-priority policy rather than a universal guarantee of safety. If the hard safety constraints and actuator limits are themselves mutually infeasible, the controller must transition to a validated emergency state.

14.3 Manual Intervention Prevention and Telemetry Stability

Rule 8.16 prohibits manual waypoint modification and manual flight-path correction during autonomous operation. The GCS architecture therefore minimizes the available low-level control surface.

Hands-Off GCS Architecture. The normal operator interface can be restricted to three high-level operations:

[LOAD MISSION FILE, START MISSION, EMERGENCY SAFETY ABORT] .

Normal waypoint dragging, joystick-based trajectory modification, and manual payload-release commands are excluded from the autonomous mission interface. This architectural restriction reduces the possibility of accidental manual intervention during mission execution.

Telemetry Interpolation for UI Stability. To maintain a temporally smooth visualization despite irregular RF packet arrival, the GCS reconstructs the displayed position using the most recently received state.

Let

$$t_{\mathrm{rx}}$$

denote the timestamp of the latest valid packet, with corresponding position, velocity, and acceleration

$$p_{\mathrm{rx}}, \qquad v_{\mathrm{rx}}, \qquad a_{\mathrm{rx}}.$$

A constant-acceleration display predictor is

$$p_{\mathrm{UI}}(t) = p_{\mathrm{rx}} + v_{\mathrm{rx}}(t - t_{\mathrm{rx}}) + \frac{1}{2} a_{\mathrm{rx}}(t - t_{\mathrm{rx}})^2.$$

The displayed state is therefore interpolated or short-horizon extrapolated locally rather than being updated only when a new RF packet arrives. This reduces visual discontinuities in the swarm map and provides a temporally smoother operator interface during transient communication degradation.

The prediction is used strictly for visualization and should not replace the onboard estimator or the safety-critical state used by the NMPC and CBF layers.

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15 GCS INTEGRATION, TELEMETRY MATHEMATICS & NIDAR SCORING OPTIMIZATION

15.1 Single-Operator GCS Architecture and Telemetry Mathematics

Rules 8.13 and 8.14 mandate a single Ground Control Station (GCS) displaying a unified interface for all drones. Furthermore, Annexure 5 prohibits GSM/LTE/5G and Internet connectivity. All mission data are therefore transmitted through a local RF mesh, such as a 900 MHz LoRa or 5.8 GHz digital telemetry system.

Transmitting raw 13-dimensional state vectors, LiDAR point clouds, and four simultaneous HD video streams over a finite-bandwidth local RF link can produce packet loss and interface latency. The GCS therefore implements a priority-based data multiplexing strategy.

15.1.1 Telemetry Bandwidth Optimization

Let

$$R_{\text{state},i}, \qquad R_{\text{video},i}, \qquad R_{\text{map},i}$$

denote the allocated data rates for state, video, and map information from drone i, respectively. The GCS solves

$$\max_{\mathbf{R}} \sum_{i=1}^{4} \left(w_{\text{s}} R_{\text{s},i} + w_{\text{v}} R_{\text{v},i} + w_{\text{m}} R_{\text{m},i} \right) \quad \text{s.t.} \quad \sum_{i=1}^{4} \left(R_{\text{s},i} + R_{\text{v},i} + R_{\text{m},i} \right) \le B_{\text{RF}}, \; R_{\text{s},i}, R_{\text{v},i}, R_{\text{m},i} \ge 0.$$

The state stream has the highest priority and is transmitted at approximately

$$f_{\text{state}} = 20\,\text{Hz}.$$

Each state packet contains the vehicle state x_i, battery information, and CBF status.

The map stream is assigned intermediate priority and is transmitted at approximately

$$f_{\text{map}} = 1\,\text{Hz}.$$

It contains compressed TSDF or terrain-map updates.

The video stream has the lowest communication priority. H.265 hardware encoding with Region-of-Interest (ROI) compression is employed. When a drone detects a potential survivor or performs a payload-delivery operation, the GCS dynamically increases the corresponding video bitrate while reducing the bitrate and/or frame rate of the remaining overview feeds.

15.1.2 UI Interpolation for Operator Confidence

To reduce visual discontinuities caused by irregular RF packet arrival, the GCS maintains a local timestamp-based state reconstruction process.

Let

$$t_{\text{rx}}$$

denote the timestamp of the latest received telemetry packet, with corresponding position, velocity, and acceleration

$$p_{\text{rx}}, \qquad v_{\text{rx}}, \qquad a_{\text{rx}}.$$

The displayed position is predicted according to

$$p_{\text{UI}}(t) = p_{\text{rx}} + v_{\text{rx}} \left(t - t_{\text{rx}} \right) + \frac{1}{2} a_{\text{rx}} \left(t - t_{\text{rx}} \right)^2 .$$

This prediction is used for visualization only and does not replace the onboard state estimator used by the NMPC or CBF safety layers. It allows the unified GCS interface to maintain a temporally smooth representation of the swarm even when telemetry packets arrive at irregular intervals.

15.2 Algorithmic Optimization for NIDAR Scoring Metrics

The mission scoring framework is mapped to explicit perception, control, search-time, and safety objectives. The resulting architecture is designed to maximize mission performance subject to the physical and computational constraints of the system.

15.2.1 Survivor Detection and Geotagging

For each detected survivor, let

$$p_{\text{surv}} \in R^3$$

denote the estimated three-dimensional position and let

$$\Sigma_{\text{surv}}$$

denote the associated covariance matrix.

The GCS promotes a detection to a confirmed survivor tag only when

$$Tr\left(\Sigma_{\text{surv}}\right) \leq \varepsilon_{\text{tag}}^2.$$

If

$$Tr\left(\Sigma_{\text{surv}}\right) > \varepsilon_{\text{tag}}^2,$$

the detection remains provisional and the corresponding drone may perform an additional observation or loiter maneuver. This reduces the probability of committing a poorly localized detection to the global mission state.

The threshold ε_{tag} should be selected consistently with the required geolocation tolerance and the units of the covariance representation.

15.2.2 Survivor Kit Delivery Accuracy

The terminal delivery objective is to minimize the position and velocity errors at payload release:

$$\|p(t_{\text{drop}}) - p_{\text{surv}}\| \to 0,$$

$$\|v(t_{\text{drop}})\| \to 0.$$

When the drone enters a terminal approach region,

$$\|p - p_{\text{surv}}\| \leq 3\,\text{m},$$

control can transition from the nominal NMPC trajectory tracker to an Image-Based Visual Servoing (IBVS) controller.

Let

$$s \in R^m$$

denote the measured image feature vector and

$$s^*$$

the desired image feature configuration. The visual-servoing control law is

$$u_{\mathrm{IBVS}} = \lambda J_v^{\dagger} \left(s^* - s \right),$$

where

$$J_v$$

is the image Jacobian,

$$J_v^{\dagger}$$

is its generalized inverse, and

$$\lambda > 0$$

is the controller gain.

The desired image feature s^* can be selected so that the survivor target is positioned at the desired pixel location, typically near the image center.

The achievable delivery accuracy depends on camera calibration, target detection error, depth estimation, vehicle dynamics, wind disturbance, estimator uncertainty, and payload-release dynamics. Therefore, a bound such as

$$\| p(t_{\mathrm{drop}}) - p_{\mathrm{surv}} \| < 0.3\,\mathrm{m}$$

should be treated as a design target until experimentally validated.

15.2.3 Fast Completion and Search-Time Optimization

The fast-completion objective requires efficient coverage of the

$$A_{\mathrm{mission}} = 100\,000\,\mathrm{m}^2$$

operational region.

Let

$$W_{\mathrm{FOV}}$$

denote the effective ground footprint width of the camera and

$$\varepsilon_{\mathrm{overlap}}$$

the prescribed fractional overlap between adjacent sweep tracks.

The nominal sweep velocity is estimated from the camera integration time t_{int} and frame rate f_{frame} as

$$v_{\mathrm{sweep}} = \frac{W_{\mathrm{FOV}} \left(1 - \varepsilon_{\mathrm{overlap}} \right)}{t_{\mathrm{int}} f_{\mathrm{frame}}}.$$

The actual admissible velocity is additionally constrained by vehicle dynamics, perception requirements, and safety constraints:

$$v_{\text{sweep}} \leq \min\left(v_{\text{perception}}, v_{\max}, v_{\text{CBF}}\right).$$

For an idealized four-drone partition with perfectly balanced search regions and negligible transition overhead, the theoretical coverage-time lower bound is

$$T_{\min} \geq \frac{A_{\text{mission}}}{4W_{\text{FOV}}v_{\text{sweep}}}.$$

Substituting

$$A_{\text{mission}} = 100\,000\,\text{m}^2,$$

gives

$$T_{\min} \geq \frac{100\,000}{4W_{\text{FOV}}v_{\text{sweep}}}.$$

The expression represents an ideal lower bound. Boustrophedon turns, unequal Voronoi cell areas, terrain-following, survivor servicing, communication delays, and reallocation events introduce additional mission time.

A system-design target may therefore be imposed as

$$T_{\min} \leq 11\,\text{min},$$

leaving approximately 4 min of nominal margin within a 15 min fast-completion target.

15.3 Penalty Avoidance via Mathematical Guardrails

The safety architecture incorporates explicit constraints intended to reduce the probability of penalty-triggering events.

15.3.1 Launch and Landing Zone Constraint

The permitted launch and landing region is represented by the polygon

$$\mathcal{Z}_{\text{launch}} \subset R^2,$$

corresponding to the prescribed

$$12\,\text{ft} \times 12\,\text{ft}$$

area, approximately

$$3.66\,\text{m} \times 3.66\,\text{m}.$$

When Return-to-Home is triggered, the Minimum-Snap trajectory is generated subject to the terminal constraint

$$p_{xy}(t_{\text{land}}) \in \mathcal{Z}_{\text{launch}},$$

together with

$$\dot{p}(t_{\text{land}}) = 0.$$

Thus, the terminal trajectory is explicitly constrained to terminate within the designated launch/landing region.

15.3.2 Zero Manual Intervention

Rule 8.16 prohibits manual waypoint modification, flight-path correction, or manual payload-release commands during autonomous operation.

The GCS is therefore designed without normal low-level manual control interfaces. The standard operator interface consists of

[LOAD MISSION FILE, START MISSION, EMERGENCY SAFETY ABORT] .

Waypoint dragging, joystick-based trajectory modification, and manual payload-release controls are excluded from the autonomous mission interface. This architectural restriction removes an accidental manual-command pathway from the normal mission-control software.

15.3.3 Geofence Protection

The mission boundary is represented by

$$\mathcal{B} \subset R^2$$

and loaded from the organizer-provided mission file during initialization.

The corresponding geofence barrier satisfies

$$h_{\text{geo}}(x) \geq 0.$$

The geofence CBF is incorporated into the NMPC constraint hierarchy. Under nominal feasibility, the controller modifies the commanded trajectory to maintain the vehicle within the admissible region.

If a sufficiently large disturbance causes actuator saturation such that the hard safety constraints and geofence constraint cannot simultaneously be satisfied, the previously defined lexicographic policy determines the fallback behavior. Therefore, arbitrary geofence preservation cannot be guaranteed when actuator authority is insufficient.

15.4 Five-Minute Setup and Mission-File Ingestion Protocol

Annexure 3 specifies a maximum setup, configuration, and mission-file ingestion interval of

$$T_{\text{setup,max}} = 300\,\text{s}.$$

The GCS therefore executes a predefined parallelized initialization state machine immediately after system power-up.

15.4.1 Automated Pre-Flight State Machine

1. $t = 0$–$60\,\text{s}$**: Boot and sensor initialization.**

   All drones boot, initialize their IMUs and ESKF estimators, initialize LiDAR processing, and establish the local RF mesh with the GCS.

2. $t = 60$–$120\,\mathrm{s}$**: Mission-file ingestion and spatial partitioning.**

   The operator loads the organizer-provided mission boundary file. The GCS computes the initial DWVT partition and generates the corresponding Boustrophedon waypoint sequences for all four drones.

3. $t = 120$–$180\,\mathrm{s}$**: Automated safety and actuator self-test.**

   The GCS performs CBF consistency checks. Where permitted by the hardware test procedure, the propulsion system is operated at a controlled low-thrust level to verify ESC calibration and the payload-release TTL interface.

4. $t = 180$–$240\,\mathrm{s}$**: State-estimation validation.**

   The ESKF uncertainty is checked against the prescribed threshold:

   $$Tr\left(P_{\mathrm{pos}}\right) \leq \varepsilon_{\mathrm{lidar}}.$$

   GNSS availability is also verified where applicable. The GCS then confirms that all four vehicles satisfy the required pre-flight conditions.

5. $t = 240\,\mathrm{s}$**: Mission authorization.**

   Once all required pre-flight conditions are valid, the operator initiates autonomous execution using

   START MISSION.

The remaining interval before the 300 s setup limit provides operational margin for initialization delays and final verification.